# MACHINE LEARNING - I

PROJECT REPORT

**ANNAPOORNA S**

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## Contents



## List of Figures

## List of Tables

# MACHINE LEARNING – I

# INN HOTEL PROJECT REPORT

## Problem Statement

A significant number of hotel bookings are called off due to cancellations or no-shows. The typical reasons for cancellations include change of plans, scheduling conflicts, etc. This is often made easier by the option to do so free of charge or preferably at a low cost, which is beneficial to hotel guests, but it is a less desirable and possibly revenue-diminishing factor for hotels to deal with. Such losses are particularly high on last-minute cancellations.

The new technologies involving online booking channels have dramatically changed customers' booking possibilities and behavior. This adds a further dimension to the challenge of how hotels handle cancellations, which are no longer limited to traditional booking and guest characteristics.

The cancellation of bookings impacts a hotel on various fronts:
1. Loss of resources (revenue) when the hotel cannot resell the room.
2. Additional costs of distribution channels by increasing commissions or paying for publicity to help sell these rooms.
3. Lowering prices last minute, so the hotel can resell a room, resulting in a reduction of the profit margin.
4. Human resources to make arrangements for the guests.

**Objective:**

The increasing number of cancellations calls for a Machine Learning based solution that can help in predicting which booking is likely to be canceled. INN Hotels Group has a chain of hotels in Portugal, and they are facing problems with the high number of booking cancellations and have reached out to your firm for data-driven solutions. As a data scientist have to analyze the data provided to find which factors have a high influence on booking cancellations, build a predictive model that can predict which bookings are going to be canceled in advance, and help in formulating profitable policies for cancellations and refunds.

**Data Description:**

The data contains the different attributes of customers' booking details. The detailed data dictionary is given below.

***Data Dictionary:***

- Booking_ID: the unique identifier of each booking
- no_of_adults: Number of adults
- no_of_children: Number of Children

- no_of_weekend_nights: Number of weekend nights (Saturday or Sunday) the guest stayed or booked to stay at the hotel
- no_of_week_nights: Number of weeknights (Monday to Friday) the guest stayed or booked to stay at the hotel
- type_of_meal_plan: Type of meal plan booked by the customer:
  - Not Selected – No meal plan selected
  - Meal Plan 1 – Breakfast
  - Meal Plan 2 – Half board (breakfast and one other meal)
  - Meal Plan 3 – Full board (breakfast, lunch, and dinner)
- required_car_parking_space: Does the customer require a car parking space? (0 - No, 1- Yes)
- room_type_reserved: Type of room reserved by the customer. The values are ciphered (encoded) by INN Hotels Group
- lead_time: Number of days between the date of booking and the arrival date
- arrival_year: Year of arrival date
- arrival_month: Month of arrival date
- arrival_date: Date of the month
- market_segment_type: Market segment designation.
- repeated_guest: Is the customer a repeated guest? (0 - No, 1- Yes)
- no_of_previous_cancellations: Number of previous bookings that were canceled by the customer prior to the current booking
- no_of_previous_bookings_not_canceled: Number of previous bookings not canceled by the customer prior to the current booking
- avg_price_per_room: Average price per day of the reservation; prices of the rooms are dynamic. (in euros)
- no_of_special_requests: Total number of special requests made by the customer (e.g. high floor, view from the room, etc)
- booking_status: Flag indicating if the booking was canceled or not.

**EDA Questions:**

1. What are the busiest months in the hotel?
2. Which market segment do most of the guests come from?
3. Hotel rates are dynamic and change according to demand and customer demographics. What are the differences in room prices in different market segments?
4. What percentage of bookings are canceled?
5. Repeating guests are the guests who stay in the hotel often and are important to brand equity. What percentage of repeating guests cancel?
6. Many guests have special requirements when booking a hotel room. Do these requirements affect booking cancellation?

## Exploratory data analysis

### Univariate analysis

This will involve creating histograms for the numerical columns to visualize their distributions, and count plots for the categorical columns to understand the frequency of each category.

**Numerical Features:**

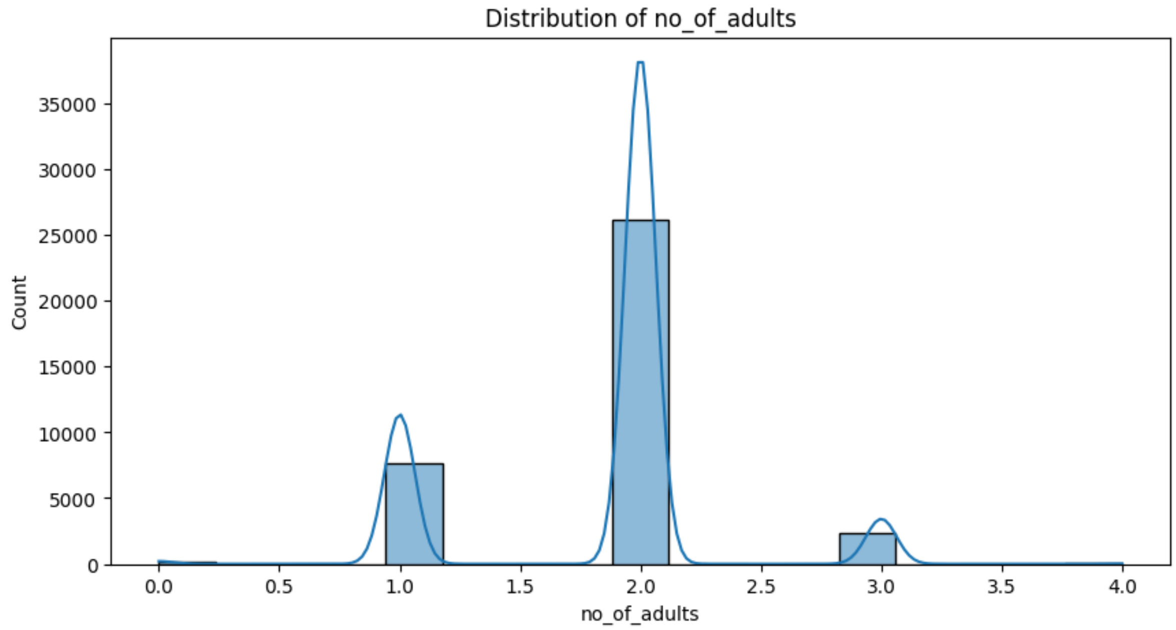


Figure 1: Distribution of number of adults

**no_of_adults**: The vast majority of bookings are for 2 adults. There are also a significant number of bookings for 1 adult, and a smaller number for 3 adults. Bookings for 0 or 4 adults are rare.

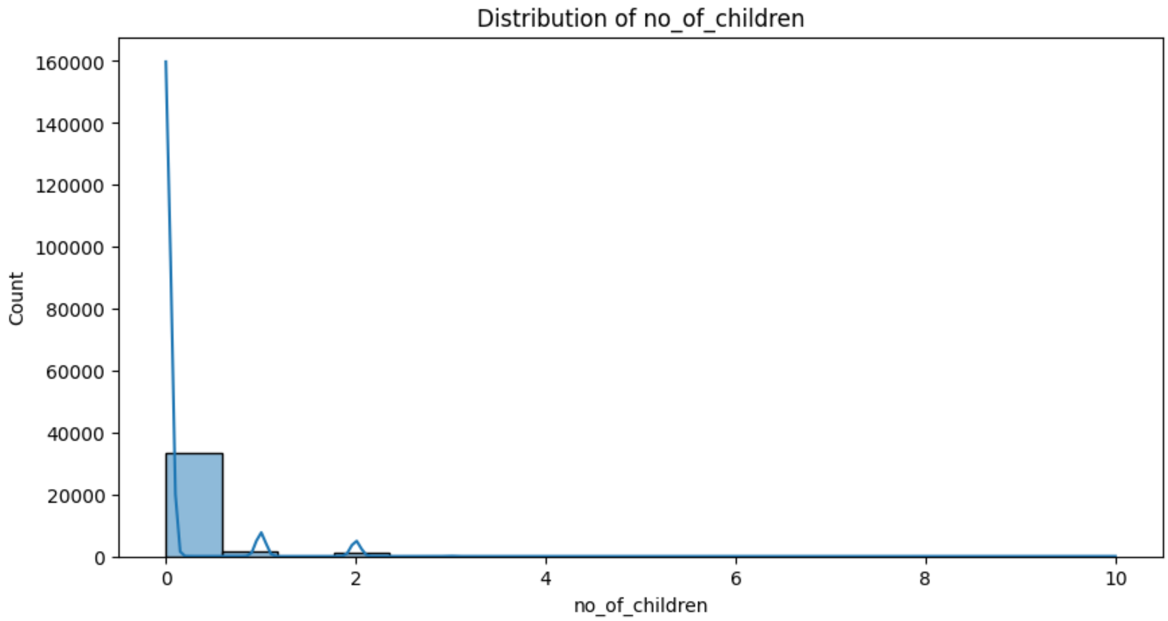


Figure 2: Distribution of number of children

**no_of_children**: Most bookings do not include children.

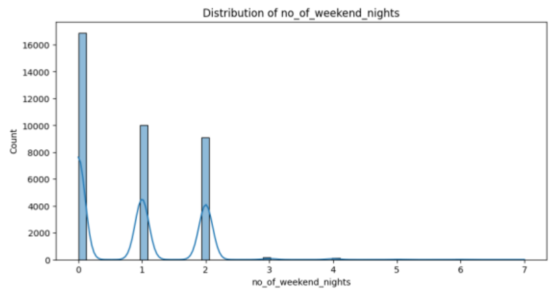


Figure 3: Distribution of no. of weekend nights

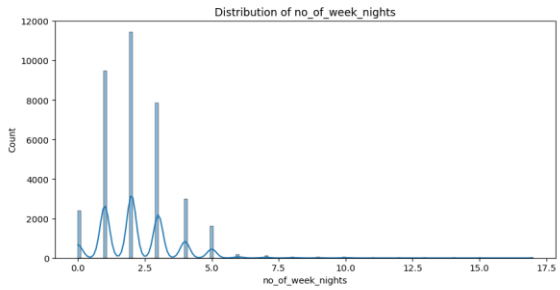


Figure 4: Distribution of no. of week nights

**no_of_weekend_nights and no_of_week_nights**: The number of weekend and week nights are skewed towards a smaller number of nights. Most guests stay for 0, 1, or 2 weekend nights, and 1, 2, or 3 week nights.

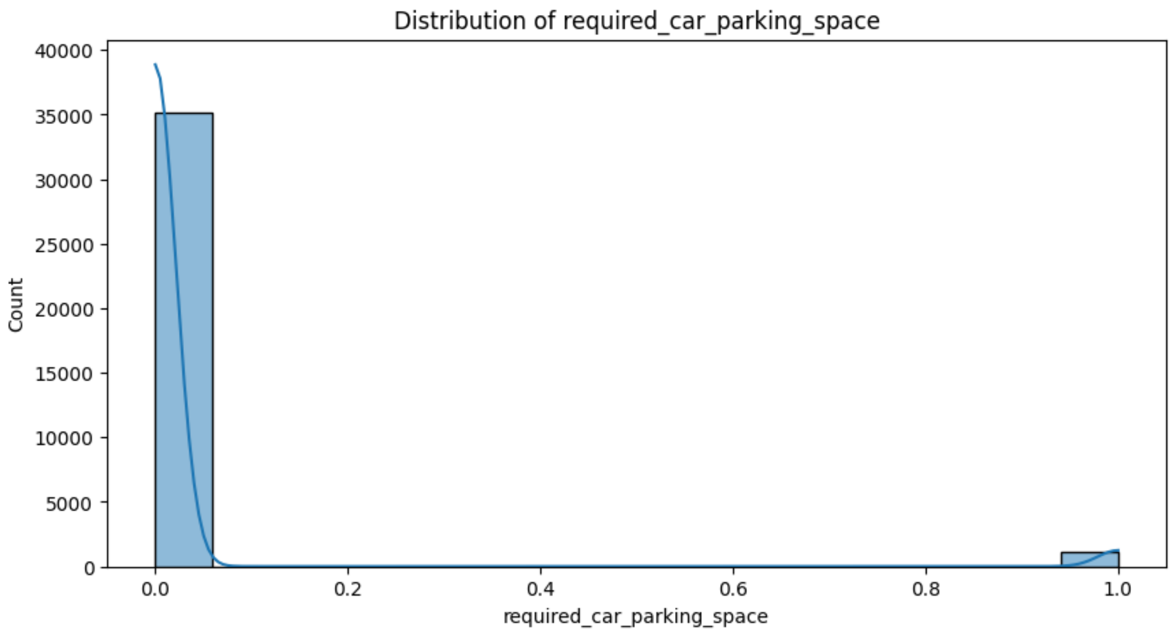


Figure 5: Distribution of required car parking space

**required_car_parking_space**: The vast majority of guests do not require a car parking space.

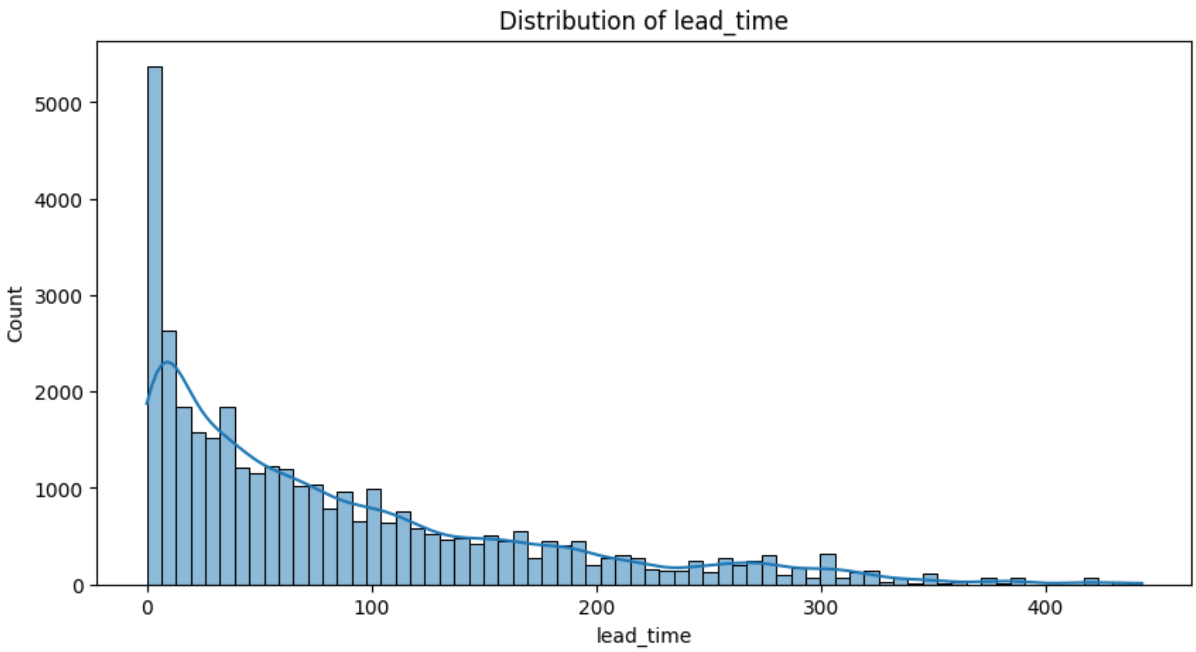


Figure 6: Distribution of  lead_time

**lead_time**: The distribution of lead time is right-skewed, indicating that most bookings are made close to the arrival date.

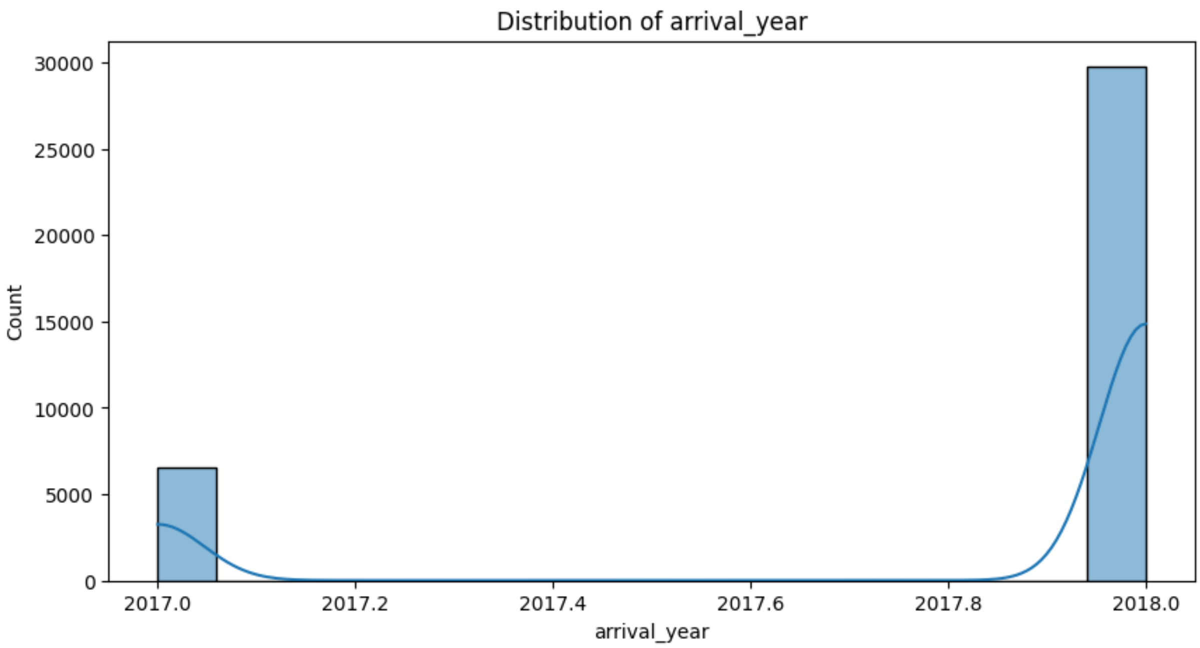


Figure 7: Distribution of arrival year

**arrival_year**: The data is heavily concentrated in the year 2018, with a smaller portion from 2017.

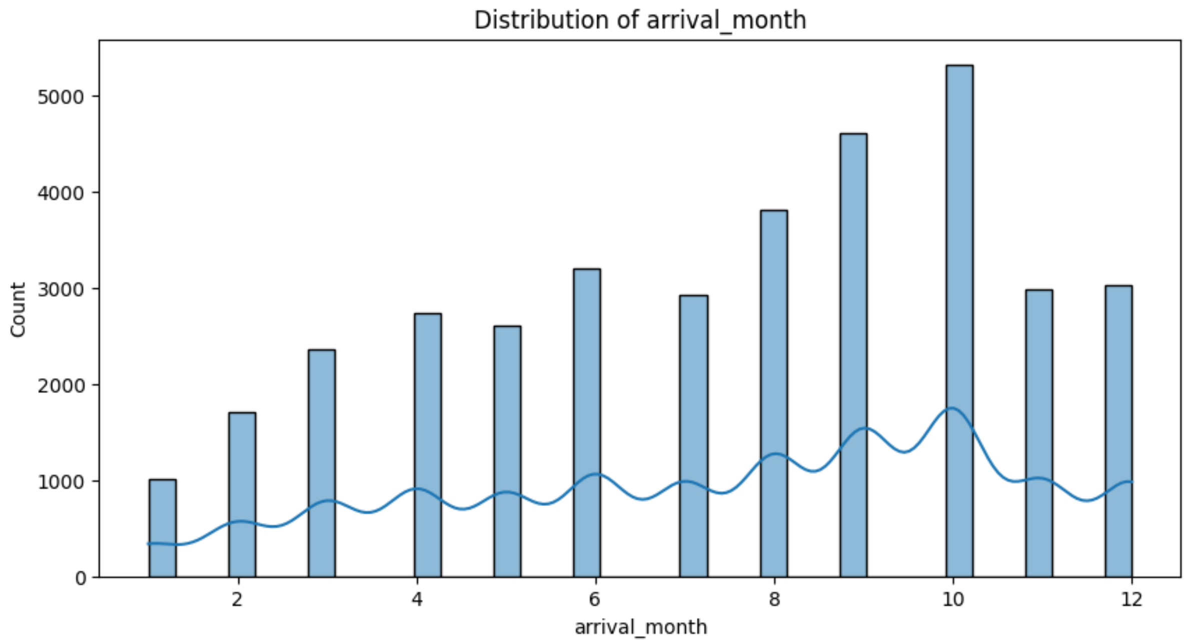


Figure 8: Distribution of arrival month

**arrival_month**: Bookings are distributed throughout the year, with a peak in the later months, especially October.

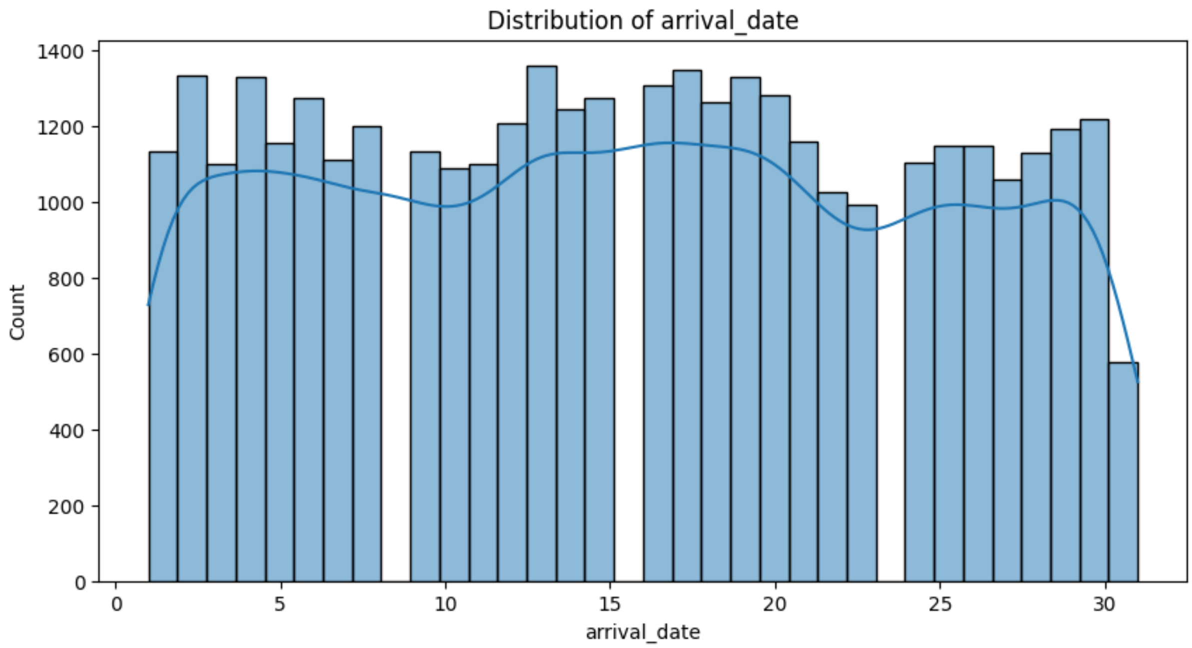


Figure 9: Distribution of arrival date

**arrival_date**: The arrival date is fairly evenly distributed throughout the month.

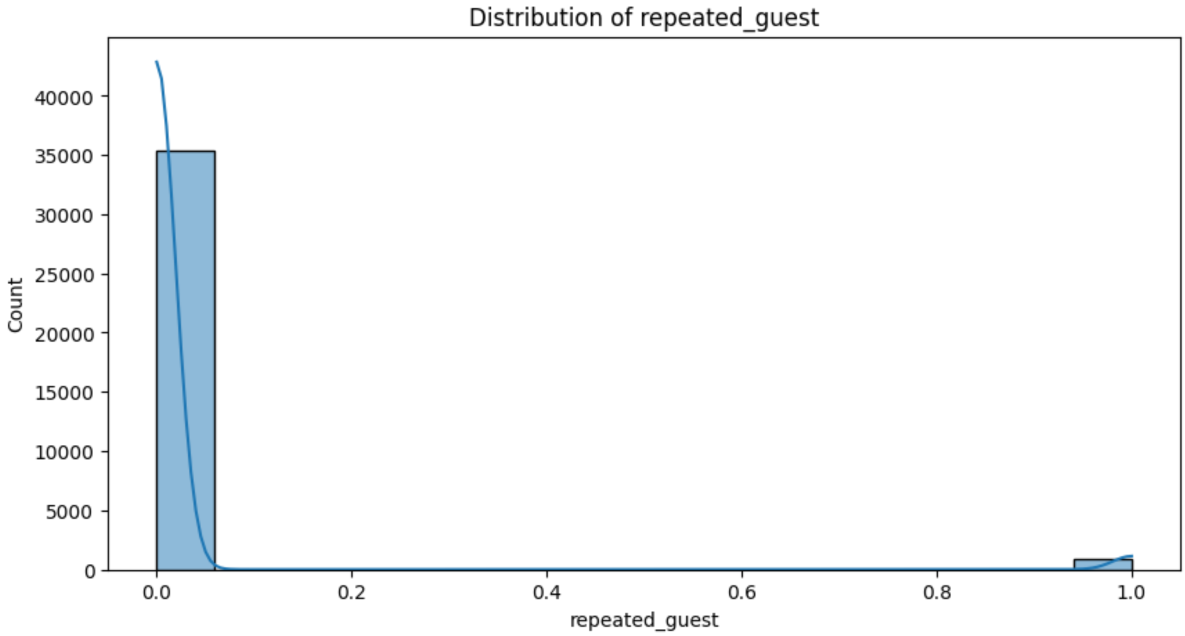


Figure 10: Distribution of repeated guest

**repeated_guest**: The overwhelming majority of guests are not repeated guests.

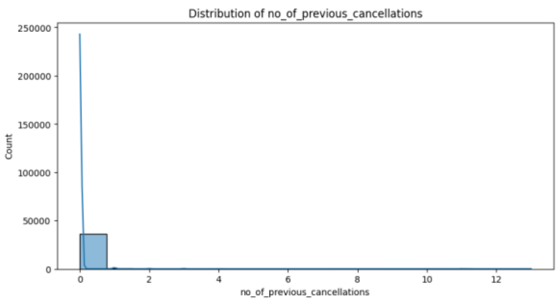


Figure 11: Distribution of no. of previous cancellations

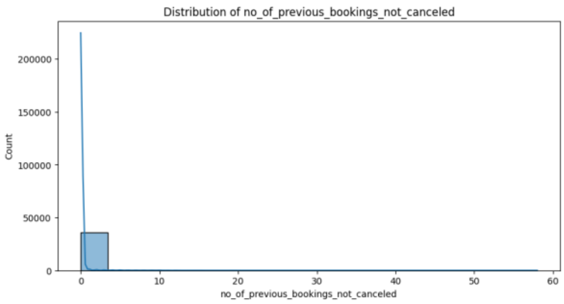


Figure 12: Distribution no. of previous bookings not cancelled

**no_of_previous_cancellations** and **no_of_previous_bookings_not_canceled**: Most guests have no prior cancellations or non-cancelled bookings.

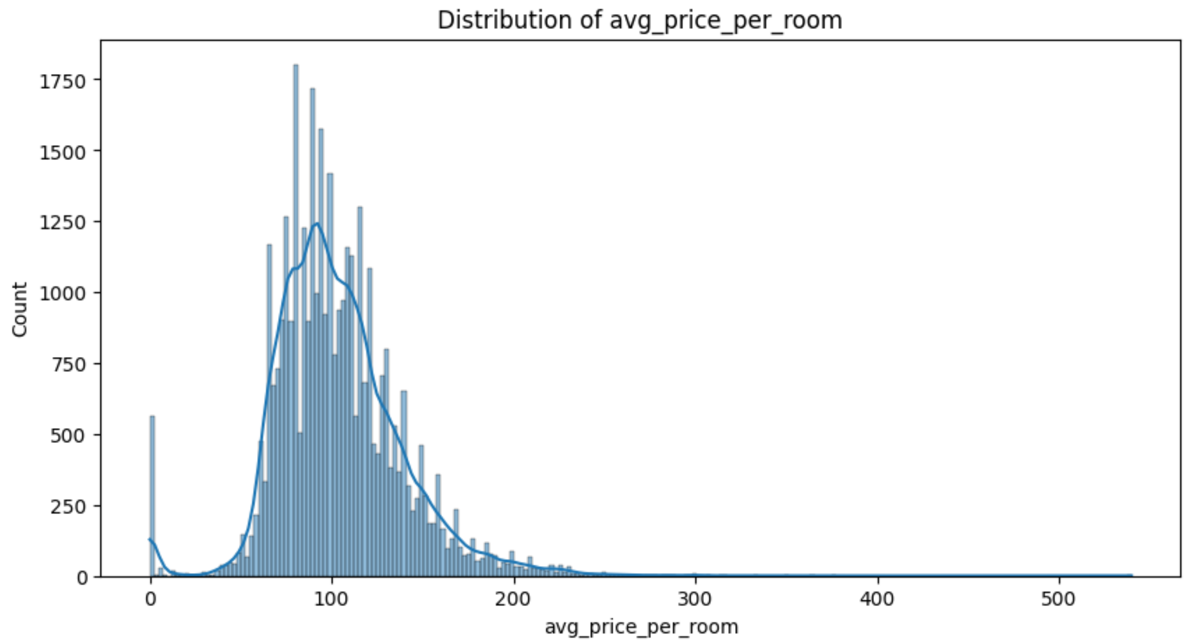


Figure 13: Distribution of avg_price per room

**avg_price_per_room**: The average price per room has a right-skewed distribution. Most rooms are priced between 50 and 150 euros. There's a small peak at 0, which might indicate free stays or data errors that need further investigation.

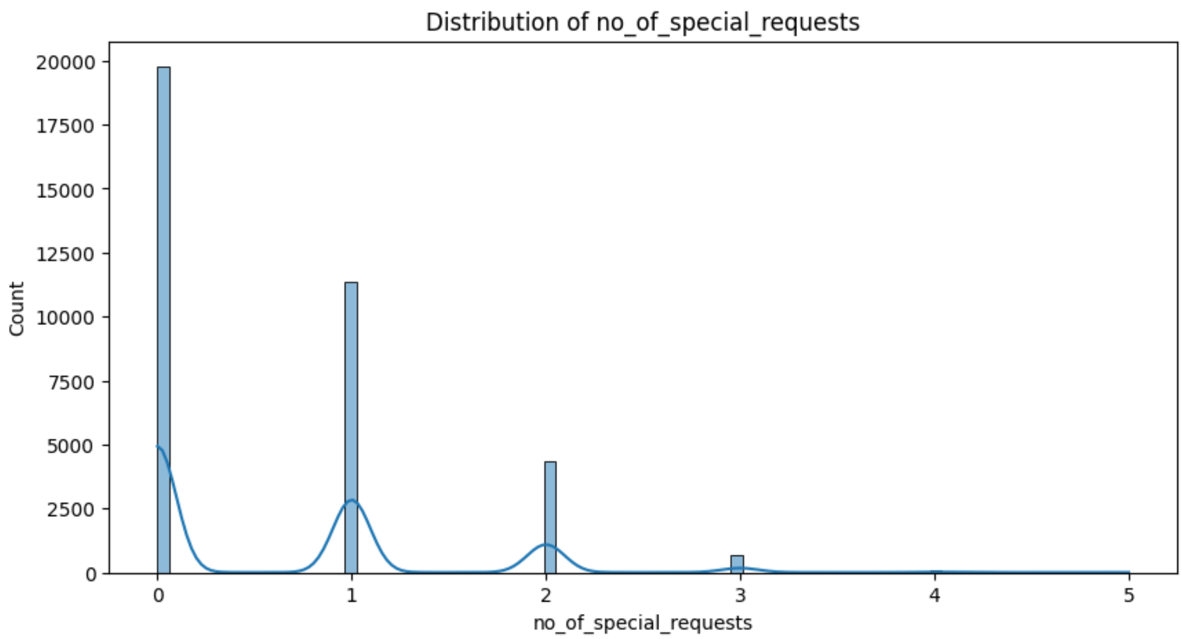


Figure 14: Distribution of number of special requests

**no_of_special_requests**: Most bookings have no special requests. A significant number have one or two requests.

**Categorical Features:**

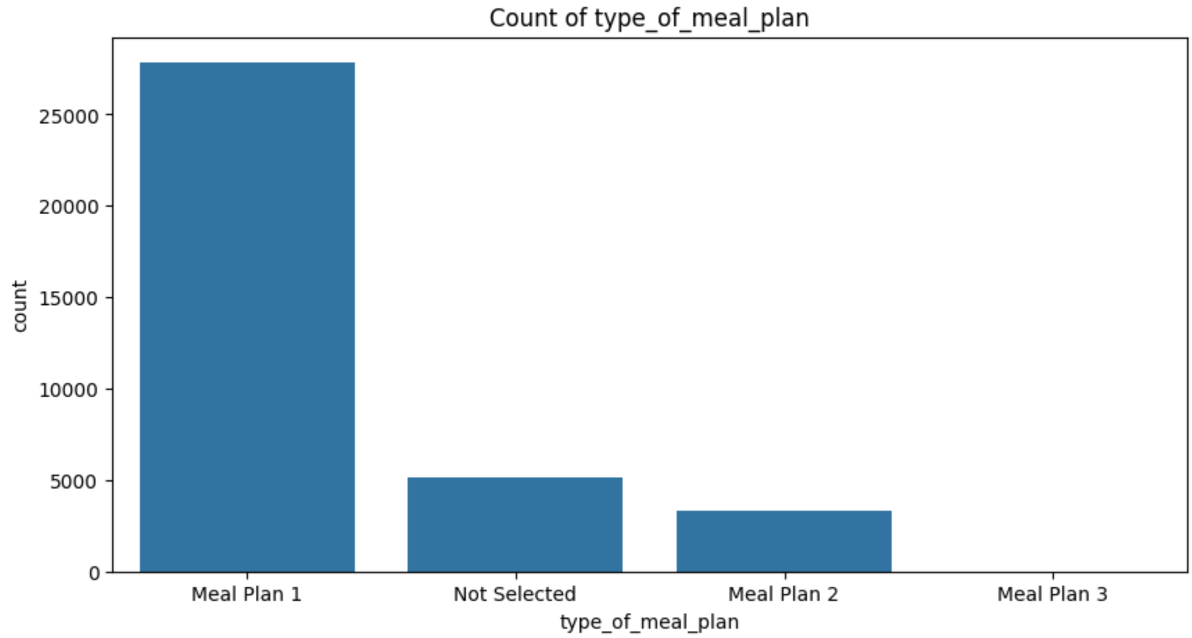


Figure 15: Count of type of meal plan

**type_of_meal_plan**: "Meal Plan 1" (Breakfast) is the most popular choice, followed by "Not Selected". "Meal Plan 2" (Half board) and "Meal Plan 3" (Full board) are less common.

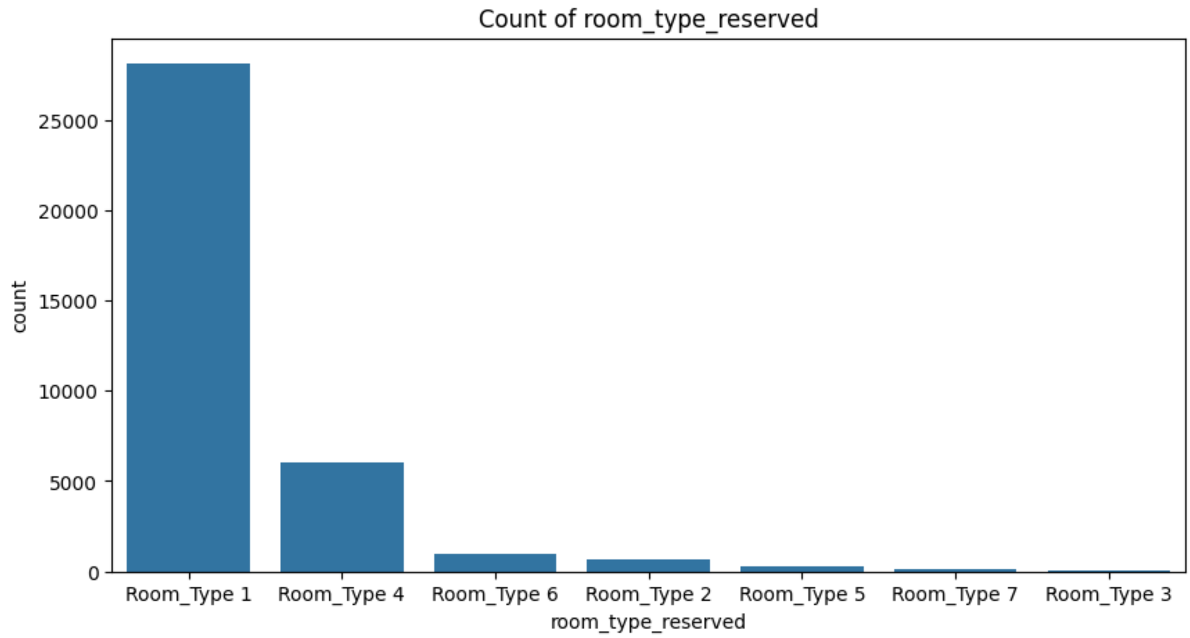


Figure 16: Count of room type reserved

**room_type_reserved**: "Room_Type 1" is by far the most commonly reserved room type.

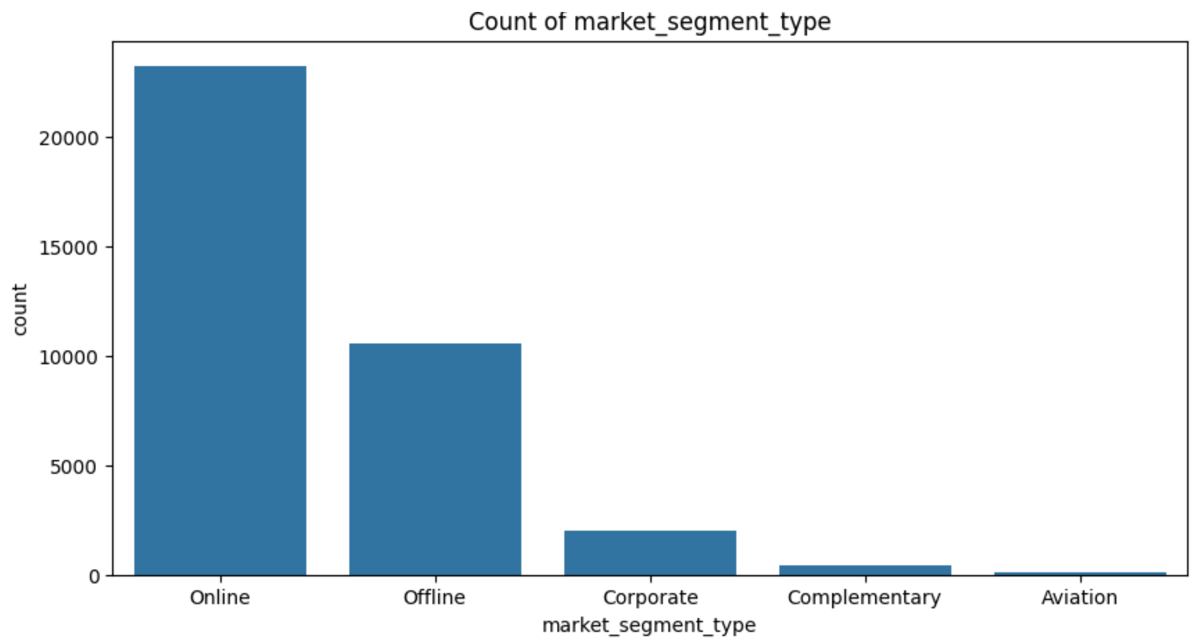


Figure 17: Count of market segment type

**market_segment_type**: The "Online" market segment is the largest, followed by "Offline".

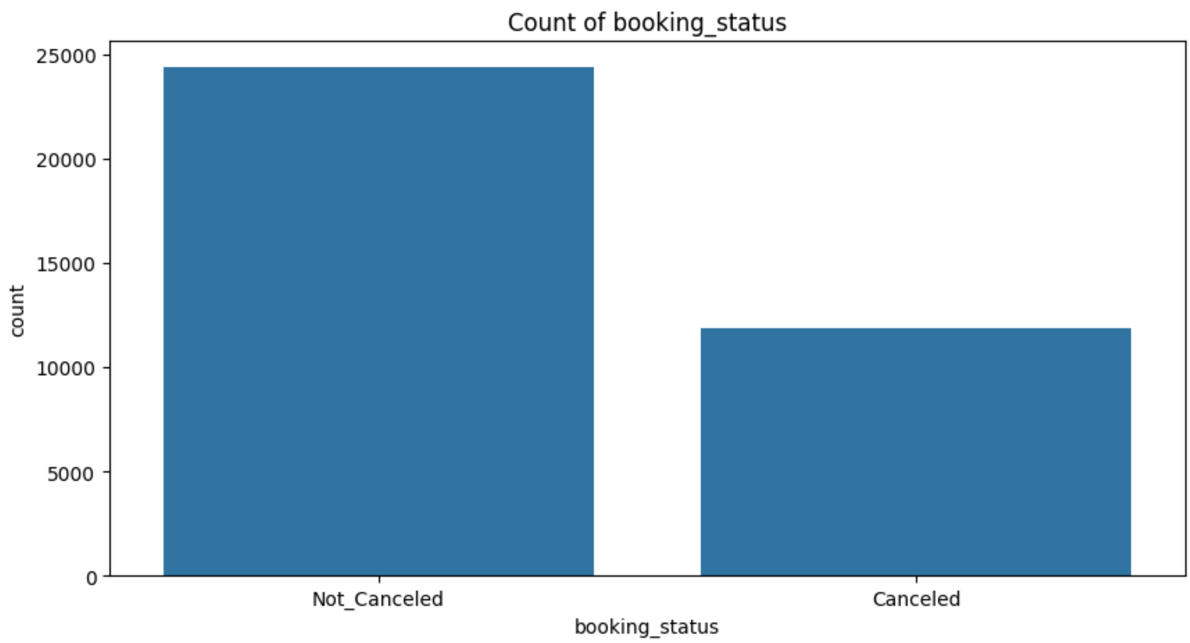


Figure 18: Count of booking status

**booking_status**: A significant portion of bookings are not canceled, but there is also a substantial number of cancellations, which highlights the importance of this analysis.

## Bivariate Analysis

The bivariate analysis has provided some interesting insights into the factors that influence booking cancellations. Here are some insights from the correlation heatmap and stacked bar plots key findings:

### Correlation Heatmap:

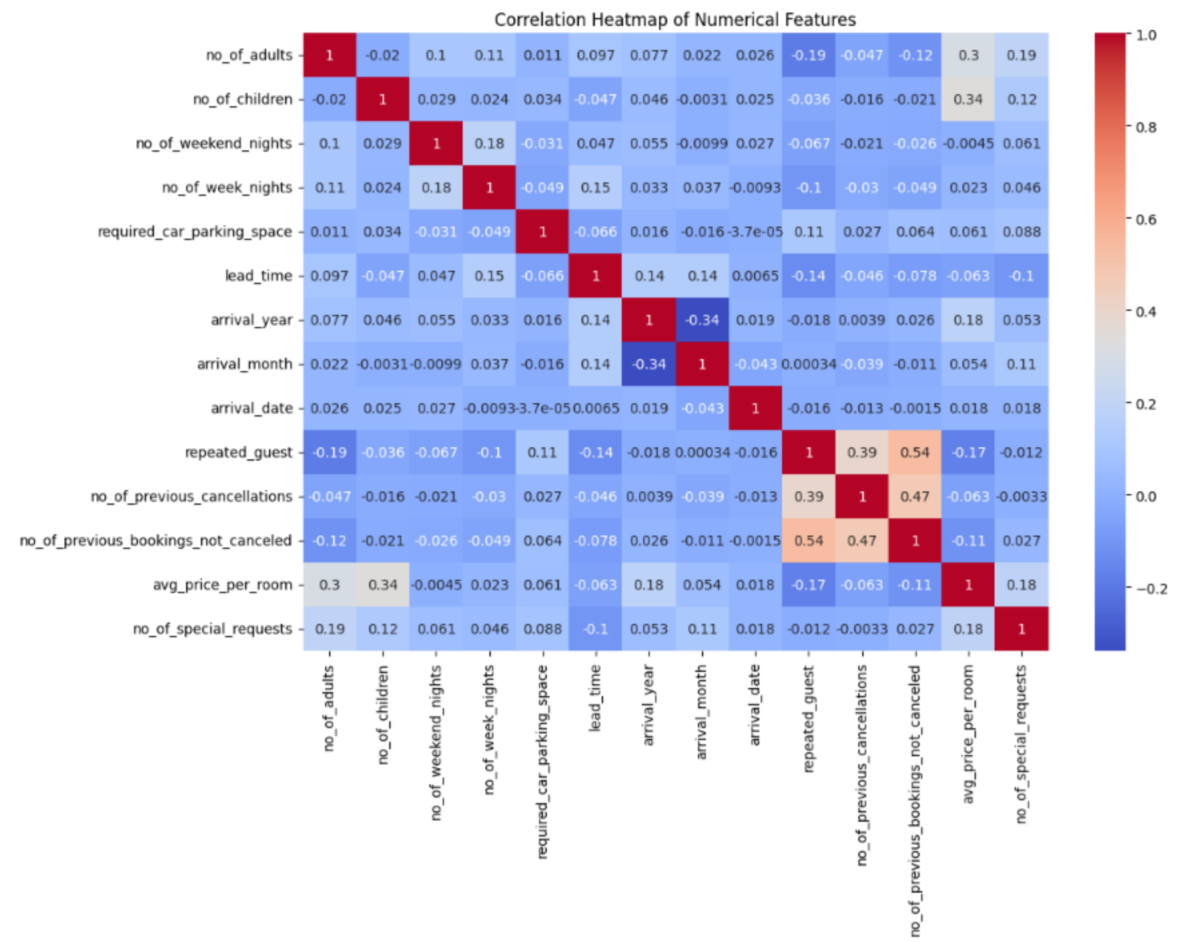


Figure 19: Correlation Heatmap of Numerical Features

- **lead_time and avg_price_per_room**: There is a moderate positive correlation (0.14) between lead_time and avg_price_per_room, suggesting that bookings made further in advance tend to have a higher average price per room.
- **no_of_previous_cancellations and repeated_guest**: There's a moderate positive correlation (0.39) between no_of_previous_cancellations and repeated_guest. This is somewhat counterintuitive and might warrant further investigation. It could be that guests who have stayed before and had to cancel are more likely to be marked as "repeated" in the system.
- **no_of_previous_bookings_not_canceled and repeated_guest**: There's a strong positive correlation (0.54)

between no_of_previous_bookings_not_canceled and repeated_guest, which is expected. Guests who have not canceled previous bookings are likely to be repeat customers.

- **arrival_year and other features**: arrival_year has a noticeable negative correlation with lead_time (-0.34) and a positive correlation with avg_price_per_room (0.16). This might indicate changes in booking patterns or pricing between 2017 and 2018.

**Stacked Bar Plot:**

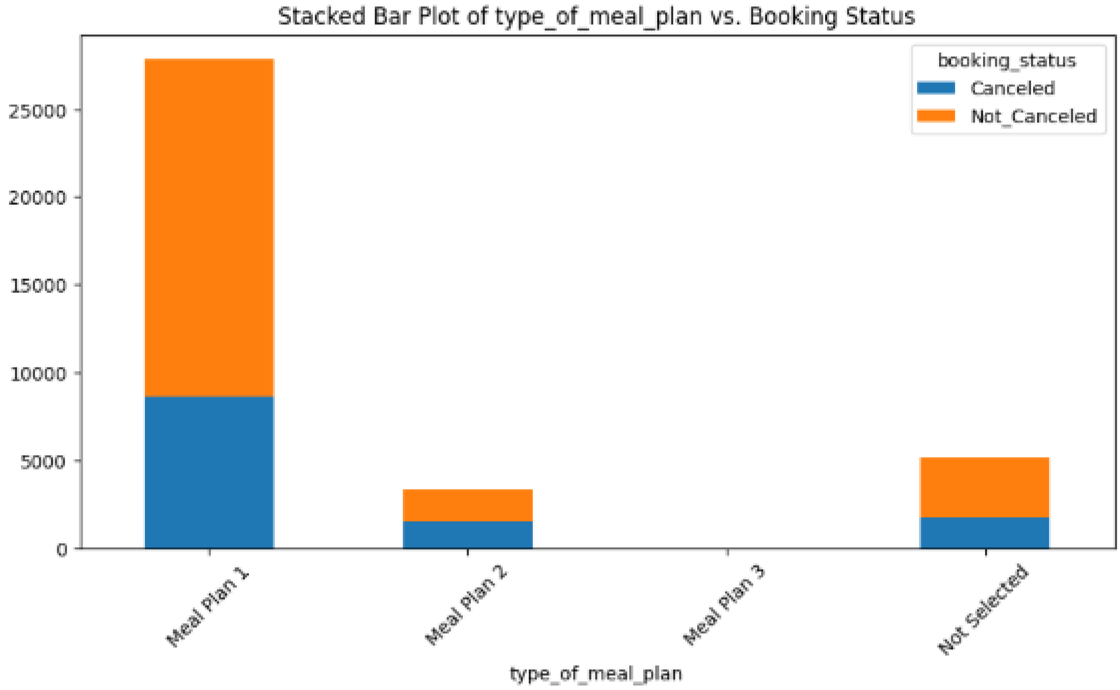


Figure 20: Stacked bar plot of type_of_meal vs. Booking status

- **type_of_meal_plan**: The *proportion* of cancellations to non-cancellations appears fairly consistent across the different meal plan types. This reinforces the earlier observation that meal plan type might not be a strong predictor of cancellation.

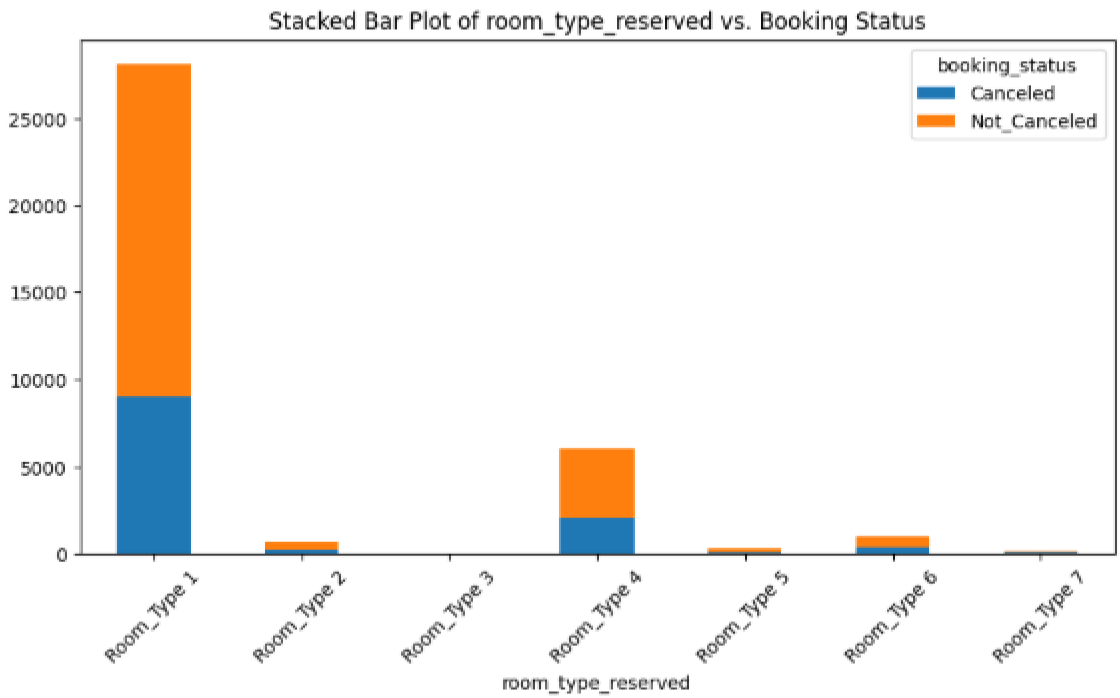


**Figure 21: Stacked bar plot of room type reserved vs. booking status**

- **room_type_reserved**: "Room_Type 1" and "Room_Type 4" have the highest number of cancellations, but this is also because they are the most booked room types. The *proportion* of cancellations seems to be highest for "Room_Type 6" and lowest for "Room_Type 2" and "Room_Type 5", although the number of bookings for these room types is small.

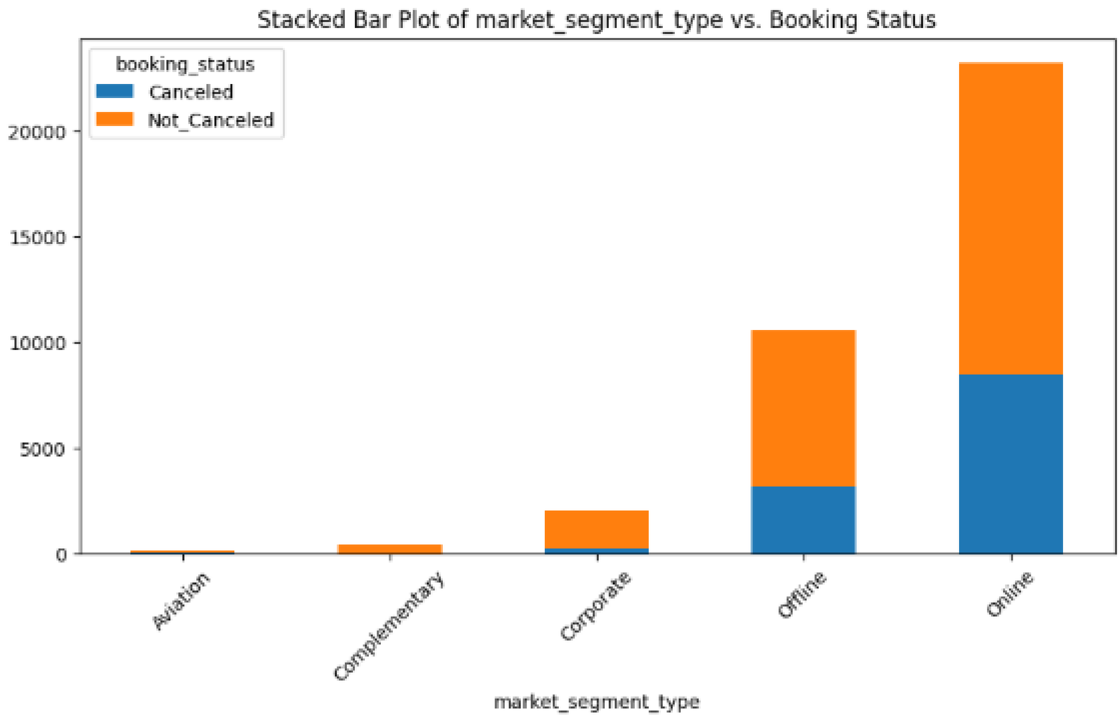


Figure 22: Stacked bar plot of market segment type vs. booking status

- **market_segment_type**: The stacked bar plot clearly shows that the "Online" market segment has a much higher proportion of cancellations compared to the "Offline" and "Corporate" segments. This is a very important finding. The "Aviation" and "Complementary" segments have a very low number of cancellations, but also a very low number of bookings overall.

- The correlation heatmap highlights some interesting relationships between numerical features, particularly around repeated_guest and past booking behavior.
- The stacked bar plots provide a clear visual representation of which categorical features are most associated with cancellations. The market_segment_type stands out as a particularly strong indicator, with online bookings being significantly more likely to be canceled.

### Use PairPlot to identify the patterns and insights

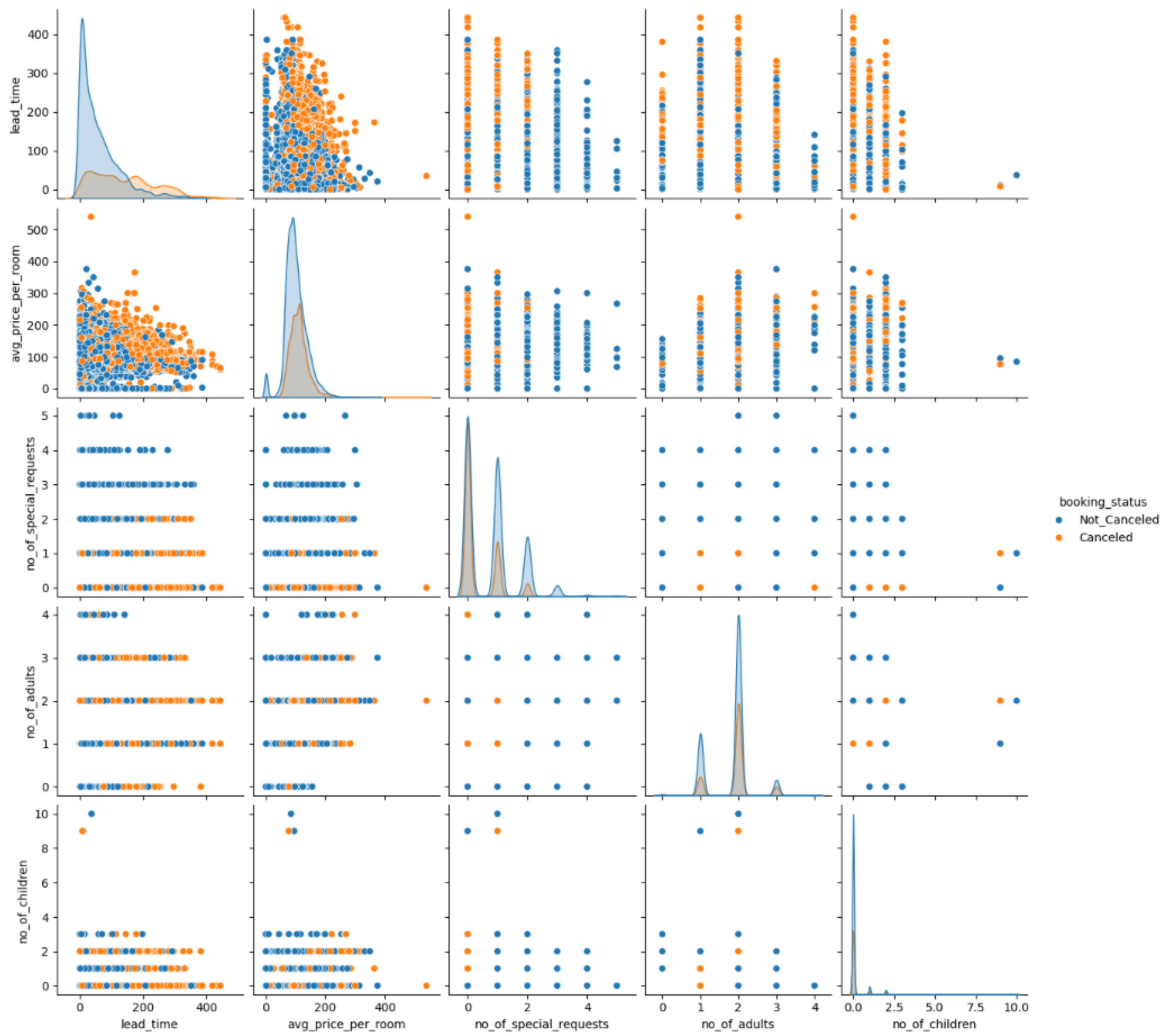


Figure 23: Pair plot of numerical features with hue as booking_status

This pairplot help us to identify any patterns or interactions between the variables that might be related to booking cancellations.

- **lead_time vs. avg_price_per_room**: There isn't a very strong linear relationship, but we can see that canceled bookings (the orange dots) are more spread out across the lead_time axis, especially for higher avg_price_per_room. This reinforces our earlier finding that longer lead times are associated with more cancellations.
- **no_of_special_requests**: The histograms on the diagonal clearly show that the distribution of no_of_special_requests is different for canceled and not-canceled bookings. Not-canceled bookings have a higher proportion of bookings with one or more special requests. The scatter plots also show that the orange dots (canceled bookings) are heavily concentrated at no_of_special_requests = 0.
- **no_of_adults and no_of_children**: The scatter plots involving no_of_adults and no_of_children don't show a clear separation between canceled and not-canceled bookings. This suggests that the number of adults and children are not strong predictors of cancellation on their own.
- **Correlations**: The pair plot visually confirms the correlations we saw in the heatmap. For example, we don't see strong linear relationships between most pairs of variables.

In summary, the pair plot reinforces the importance of lead_time and no_of_special_requests as predictors of cancellation.

**EDA Questions:**

1. **What are the busiest months in the hotel?**

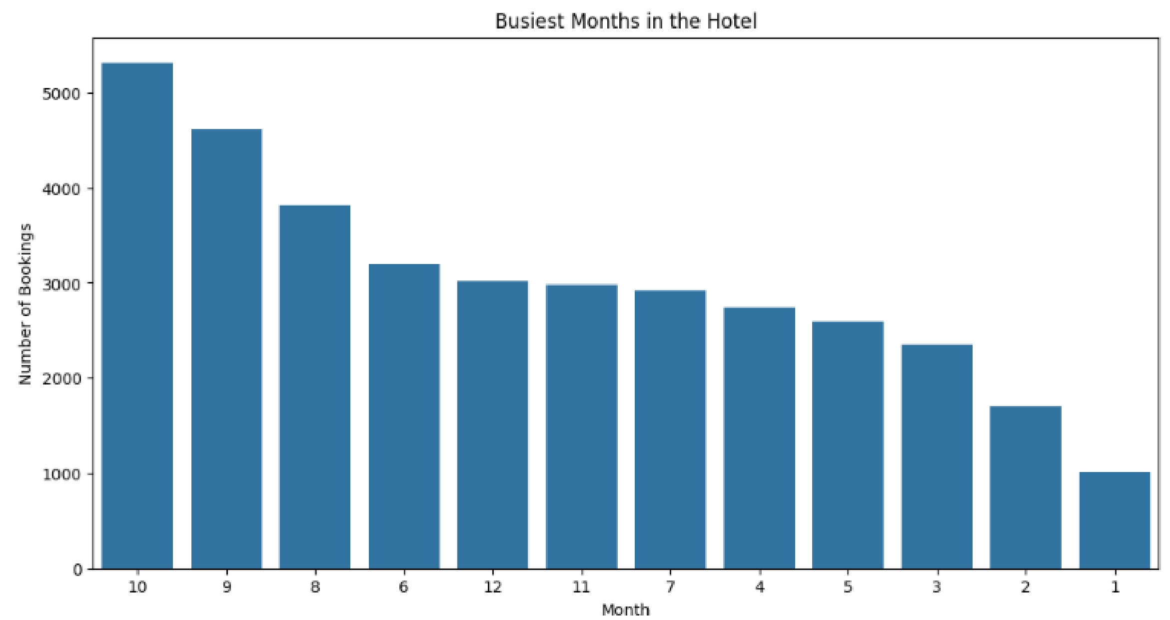


Figure 24: Busiest months in the hotel

Based on the analysis of the booking data, the busiest months for the hotel are in the later part of the year.

**top 3 busiest months are:**

1. **October**
2. **September**
3. **August**

Following these, the months of June, December, and November also show a high number of bookings. The first few months of the year, particularly January and February, are the quietest.

2. **Which market segment do most of the guests come from?**

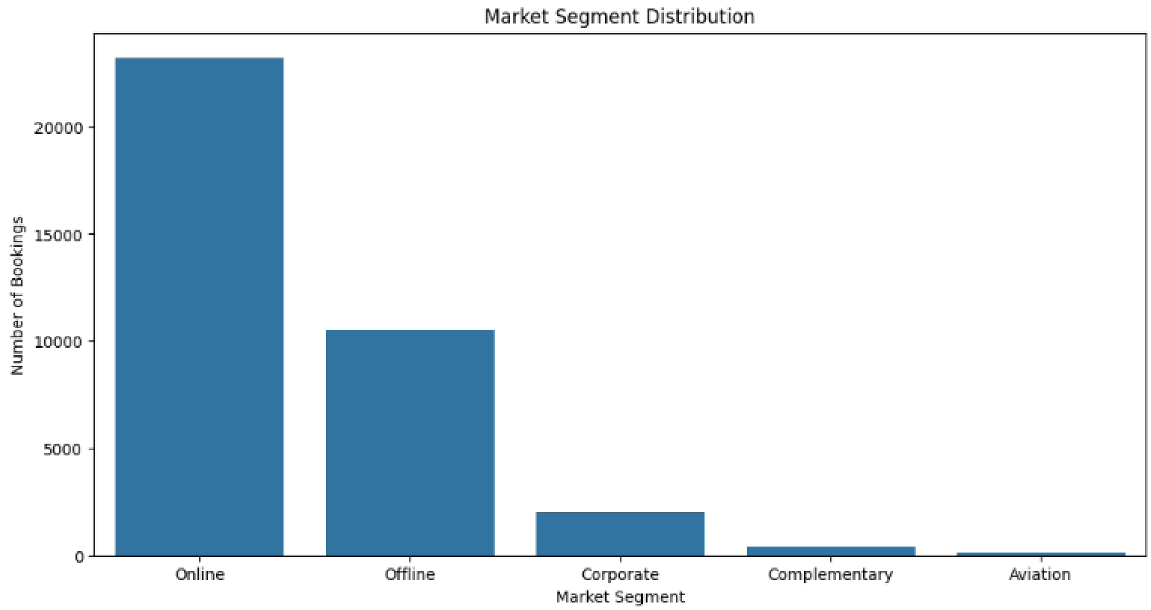


Figure 25: Market Segment distribution

The "Online" segment is significantly larger than all other segments combined, followed by the "Offline" segment. The "Corporate", "Complementary", and "Aviation" segments make up a much smaller portion of the total bookings.

3. **Hotel rates are dynamic and change according to demand and customer demographics. What are the differences in room prices in different market segments?**

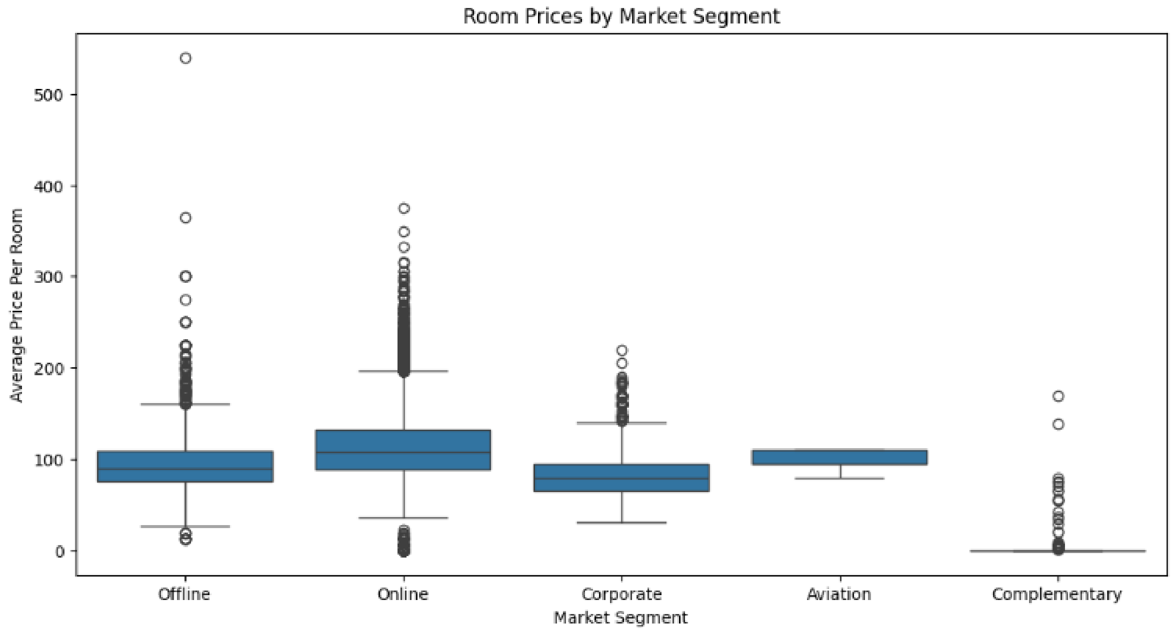


Figure 26: Room prices by Market Segment

The "Online" and "Offline" market segments have a similar median room price and a wide range of prices, with some very high-priced outliers. The "Corporate" and "Aviation" segments have a lower median price and a much smaller price range. The "Complementary" segment has the lowest prices, with most bookings being free or very cheap.

**4. What percentage of bookings are canceled?**

```
cancellation_percentage = df['booking_status'].value_counts(normalize=True) * 100
```

Percentage of bookings canceled:
booking_status
Not_Canceled    67.236389
Canceled        32.763611

**5. Repeating guests are the guests who stay in the hotel often and are important to brand equity. What percentage of repeating guests cancel?**

Let **C** be the set of all bookings.

Let **R** be the subset of bookings made by repeating guests, where **R** ⊂ **C**.

Let **Cancelled(S)** be the number of canceled bookings in a set of bookings **S**.

Let **Total(S)** be the total number of bookings in a set of bookings **S**.

The percentage of cancellations for repeating guests can be calculated as

Percentage of Cancellations (Repeating Guests) = (Cancelled(R) / Total(R)) * 100

Percentage of cancellations for repeating guests:
booking_status
Not_Canceled    98.27957
Canceled        1.72043

6. **Many guests have special requirements when booking a hotel room. Do these requirements affect booking cancellation?**

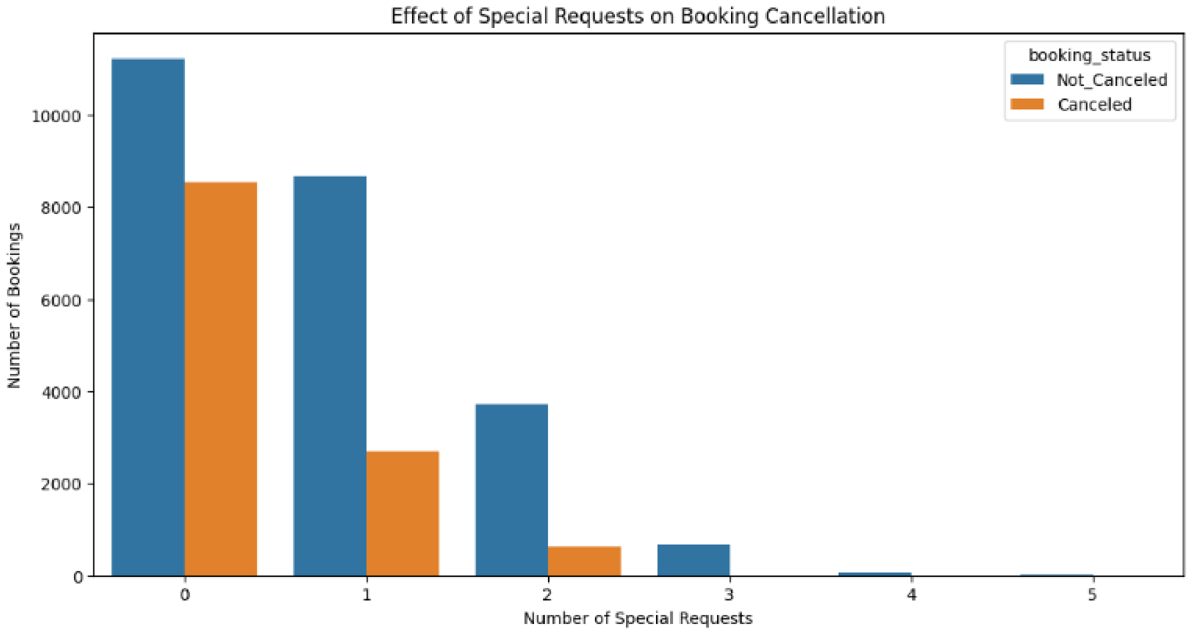


Figure 27: Effect of special requests on Booking cancellation

Yes, special requirements have a significant impact on booking cancellations. The analysis shows a clear trend: the more special requests a customer makes, the less likely they are to cancel their booking. Bookings with zero special requests have the highest number of cancellations, and this number drops sharply as the number of requests increases.

## Data Preprocessing

### Missing Value Treatment (with rationale if needed)

| | 0 |
|---|---|
| Booking_ID | 0 |
| no_of_adults | 0 |
| no_of_children | 0 |
| no_of_weekend_nights | 0 |
| no_of_week_nights | 0 |
| type_of_meal_plan | 0 |
| required_car_parking_space | 0 |
| room_type_reserved | 0 |
| lead_time | 0 |
| arrival_year | 0 |
| arrival_month | 0 |
| arrival_date | 0 |
| market_segment_type | 0 |
| repeated_guest | 0 |
| no_of_previous_cancellations | 0 |
| no_of_previous_bookings_not_canceled | 0 |
| avg_price_per_room | 0 |
| no_of_special_requests | 0 |
| booking_status | 0 |

Table 1: Count of missing values

There is no missing values in the given data set to treat

## Outlier Detection and Treatment (with rationale if needed)

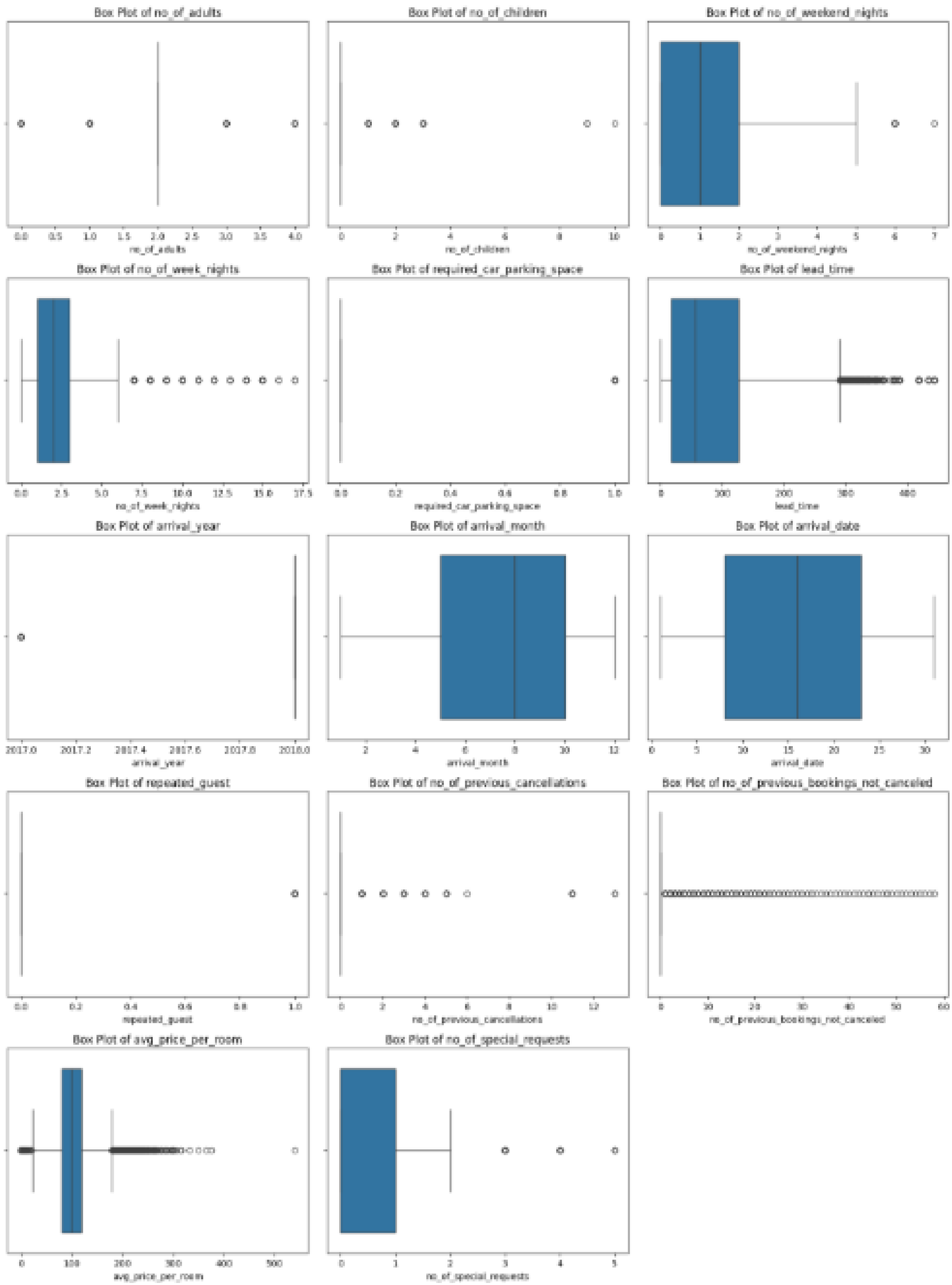


Figure 28: Boxplots to identify Identify outliers

Outliers are data points that are significantly different from other observations. They can be caused by measurement errors or by rare events. In our case, outliers in features like lead_time and avg_price_per_room could skew our analysis and negatively impact the performance of our machine learning model. Therefore, it's important to handle them appropriately.

### Outlier Treatment using the IQR Method

A common method for handling outliers is the Interquartile Range (IQR) method. We will calculate the IQR for each numerical column and then cap the outliers by replacing them with the upper and lower bounds. The upper bound is defined as Q3 + 1.5 * IQR, and the lower bound is defined as Q1 - 1.5 * IQR. Any value above the upper bound or below the lower bound will be replaced by the respective bound.

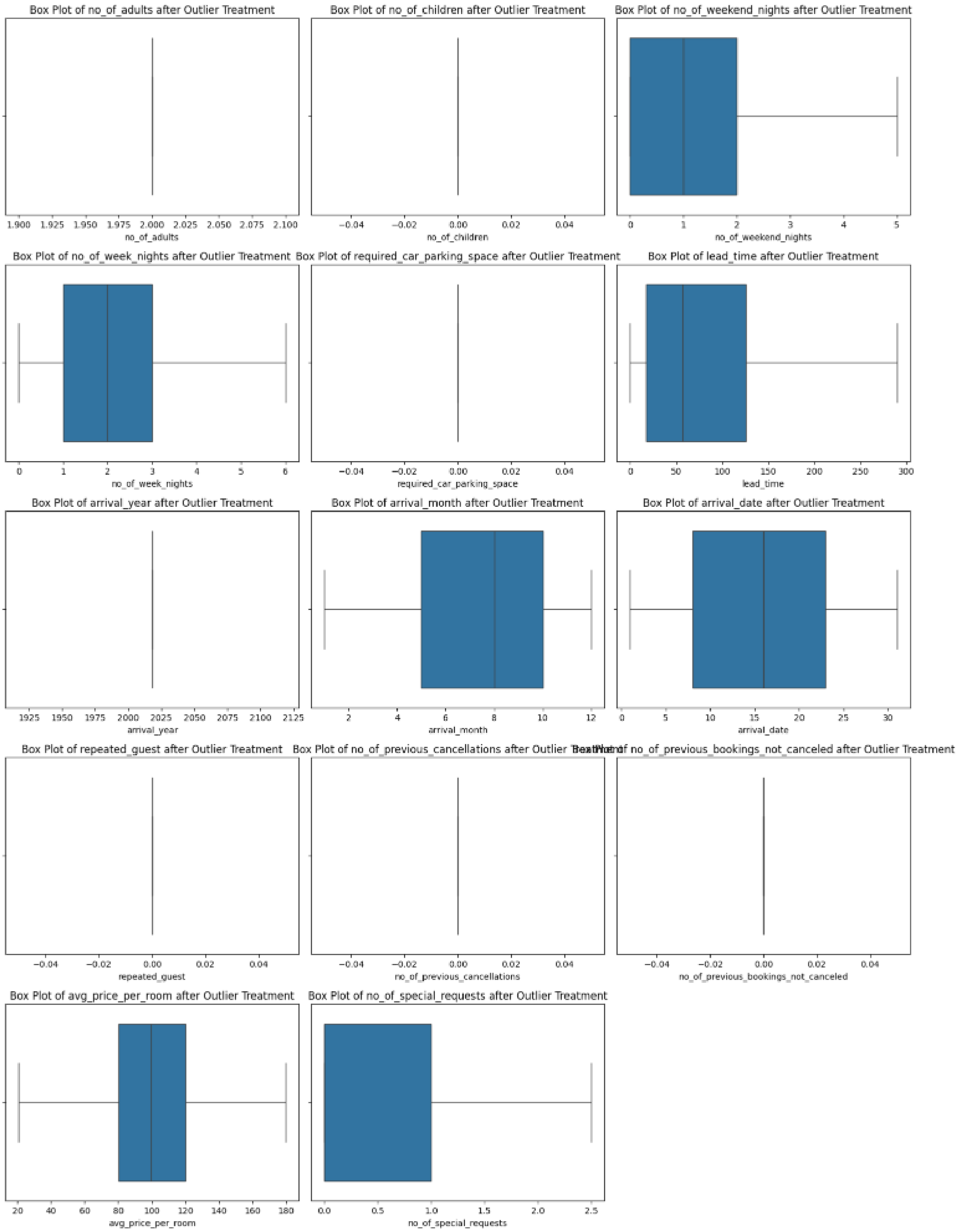


**Figure 29: Boxplot after outlier treatment**

After applying the IQR method, the extreme outliers in our numerical features have been effectively capped. This results in a more concentrated data distribution, which will help in building a more robust and reliable predictive model.

### Feature Engineering (with rationale if needed)

We have already performed some initial feature engineering by dropping the `Booking_ID`

1. **Create a total_nights feature**: We can combine no_of_weekend_nights and no_of_week_nights to get the total duration of the stay. This could be a more powerful feature than the individual night counts.
2. **Create a is_repeated feature**: The repeated_guest column is already in a binary format, but we can rename it to is_repeated for clarity.

| | no_of_adults | no_of_children | type_of_meal_plan | required_car_parking_space | room_type_reserved | lead_time | arrival_year | arrival_month | arrival_date | market_segment_type | no_of_previous_cancellations | no_of_previous_bookings_not_canceled | avg_price_per_room | no_of_special_requests | booking_status | total_nights | is_repeated |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 2 | 0 | Meal Plan 1 | 0 | Room_Type 1 | 224.0 | 2018 | 10 | 2 | Offline | 0 | 0 | 65.00 | 0.0 | Not_Canceled | 3 | 0 |
| 1 | 2 | 0 | Not Selected | 0 | Room_Type 1 | 5.0 | 2018 | 11 | 6 | Online | 0 | 0 | 106.68 | 1.0 | Not_Canceled | 5 | 0 |
| 2 | 2 | 0 | Meal Plan 1 | 0 | Room_Type 1 | 1.0 | 2018 | 2 | 28 | Online | 0 | 0 | 60.00 | 0.0 | Canceled | 3 | 0 |
| 3 | 2 | 0 | Meal Plan 1 | 0 | Room_Type 1 | 211.0 | 2018 | 5 | 20 | Online | 0 | 0 | 100.00 | 0.0 | Canceled | 2 | 0 |
| 4 | 2 | 0 | Not Selected | 0 | Room_Type 1 | 48.0 | 2018 | 4 | 11 | Online | 0 | 0 | 94.50 | 0.0 | Canceled | 2 | 0 |

Table 2: Data set after feature engineering

### Data Scaling (with rationale if needed)

Our numerical features, such as lead_time, avg_price_per_room, and total_nights, have very different scales. For example, lead_time can range from 0 to over 300, while total_nights is usually a small number.

By scaling our data, we bring all the features to a similar scale, ensuring that each feature contributes proportionally to the final prediction.

I will use the StandardScaler from scikit-learn to scale our numerical features. This will transform the data so that it has a mean of 0 and a standard deviation of 1.

| | no_of_adults | no_of_children | type_of_meal_plan | required_car_parking_space | room_type_reserved | lead_time | arrival_year | arrival_month | arrival_date | market_segment_type | no_of_previous_cancellations | no_of_previous_bookings_not_canceled | avg_price_per_room | no_of_special_requests | booking_status | total_nights | is_repeated |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.0 | 0.0 | Meal Plan 1 | 0.0 | Room_Type 1 | 1.717432 | 0.0 | 0.839242 | -1.555662 | Offline | 0.0 | 0.0 | -1.198524 | -0.812081 | Not_Canceled | 0.007353 | 0.0 |
| 1 | 0.0 | 0.0 | Not Selected | 0.0 | Room_Type 1 | -0.964558 | 0.0 | 1.164990 | -1.098013 | Online | 0.0 | 0.0 | 0.117100 | 0.526717 | Not_Canceled | 1.228051 | 0.0 |
| 2 | 0.0 | 0.0 | Meal Plan 1 | 0.0 | Room_Type 1 | -1.013544 | 0.0 | -1.766747 | 1.419055 | Online | 0.0 | 0.0 | -1.356349 | -0.812081 | Canceled | 0.007353 | 0.0 |
| 3 | 0.0 | 0.0 | Meal Plan 1 | 0.0 | Room_Type 1 | 1.558227 | 0.0 | -0.789501 | 0.503757 | Online | 0.0 | 0.0 | -0.093753 | -0.812081 | Canceled | -0.602996 | 0.0 |
| 4 | 0.0 | 0.0 | Not Selected | 0.0 | Room_Type 1 | -0.437957 | 0.0 | -1.115250 | -0.525952 | Online | 0.0 | 0.0 | -0.267360 | -0.812081 | Canceled | -0.602996 | 0.0 |

Table 3: Scale numerical data

Perform one-hot encoding on the categorical variables to convert them into a numerical format. I will also convert the booking_status column into a binary format, which will be our target variable for the machine learning model.

| | no_of_adults | no_of_children | required_car_parking_space | lead_time | arrival_year | arrival_month | arrival_date | no_of_previous_cancellations | no_of_previous_bookings_not_canceled | avg_price_per_room | ... | room_type_reserved_Room_Type 2 | room_type_reserved_Room_Type 3 | room_type_reserved_Room_Type 4 | room_type_reserved_Room_Type 5 | room_type_reserved_Room_Type 6 | room_type_reserved_Room_Type 7 | market_segment_type_Complementary | market_segment_type_Corporate | market_segment_type_Offline | market_segment_type_Online |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.0 | 0.0 | 0.0 | 1.717432 | 0.0 | 0.839242 | -1.555662 | 0.0 | 0.0 | -1.198524 | ... | False | False | False | False | False | False | False | False | True | False |
| 1 | 0.0 | 0.0 | 0.0 | -0.964558 | 0.0 | 1.164990 | -1.098013 | 0.0 | 0.0 | 0.117100 | ... | False | False | False | False | False | False | False | False | False | True |
| 2 | 0.0 | 0.0 | 0.0 | -1.013544 | 0.0 | -1.766747 | 1.419055 | 0.0 | 0.0 | -1.356349 | ... | False | False | False | False | False | False | False | False | False | True |
| 3 | 0.0 | 0.0 | 0.0 | 1.558227 | 0.0 | -0.789501 | 0.503757 | 0.0 | 0.0 | -0.093753 | ... | False | False | False | False | False | False | False | False | False | True |
| 4 | 0.0 | 0.0 | 0.0 | -0.437957 | 0.0 | -1.115250 | -0.525952 | 0.0 | 0.0 | -0.267360 | ... | False | False | False | False | False | False | False | False | False | True |

5 rows × 27 columns

Table 4: Data set after performing one hot encoding to scale

### Train-test split

I will now split the data into training and testing sets. We will use 80% of the data for training our model and the remaining 20% for testing its performance.

Now that we have our training and testing data, we are ready to build our first machine learning model.

## Model Building

### Choose the metric to optimize for the problem

The **F1-score** is the harmonic mean of precision and recall, and it provides a good balance between the two. It's a great metric to use when you want to find a balance between minimizing both false positives and false negatives.

### Build the following models

**Logistic Regression (statsmodels)**

```
                    Logit Regression Results
==============================================================================
Dep. Variable:       booking_status   No. Observations:            29020
Model:                        Logit   Df Residuals:                29000
Method:                         MLE   Df Model:                       19
Date:              Sat, 09 Aug 2025   Pseudo R-squ.:              0.3138
Time:                      14:40:24   Log-Likelihood:             -12595.
converged:                    False   LL-Null:                    -18355.
Covariance Type:          nonrobust   LLR p-value:                 0.000
==============================================================================
=====================
                                       coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------------------------
const                               -0.3327      0.222     -1.500      0.134      -0.767       0.102
lead_time                            1.3524      0.020     68.620      0.000       1.314       1.391
arrival_month                       -0.1896      0.017    -11.100      0.000      -0.223      -0.156
arrival_date                         0.0168      0.016      1.072      0.284      -0.014       0.048
avg_price_per_room                   0.6471      0.022     29.479      0.000       0.604       0.690
no_of_special_requests              -1.0954      0.021    -52.972      0.000      -1.136      -1.055
total_nights                         0.1184      0.016      7.331      0.000       0.087       0.150
type_of_meal_plan_Meal Plan 2        0.1444      0.058      2.470      0.014       0.030       0.259
type_of_meal_plan_Meal Plan 3       11.9800    259.379      0.046      0.963    -496.394     520.354
type_of_meal_plan_Not Selected       0.3260      0.049      6.682      0.000       0.230       0.422
room_type_reserved_Room_Type 2      -0.3128      0.118     -2.647      0.008      -0.544      -0.081
room_type_reserved_Room_Type 3       0.9037      1.569      0.576      0.565      -2.171       3.979
room_type_reserved_Room_Type 4      -0.2101      0.048     -4.387      0.000      -0.304      -0.116
room_type_reserved_Room_Type 5      -0.6805      0.195     -3.484      0.000      -1.063      -0.298
room_type_reserved_Room_Type 6      -0.4903      0.105     -4.685      0.000      -0.695      -0.285
room_type_reserved_Room_Type 7      -0.6248      0.271     -2.304      0.021      -1.156      -0.093
market_segment_type_Complementary  -23.8608   1215.440     -0.020      0.984   -2406.079    2358.358
market_segment_type_Corporate       -1.2492      0.238     -5.253      0.000      -1.715      -0.783
market_segment_type_Offline         -1.9334      0.225     -8.580      0.000      -2.375      -1.492
market_segment_type_Online          -0.1800      0.222     -0.810      0.418      -0.615       0.255
==============================================================================
=====================

Statsmodels Logistic Regression Performance:
```

```
            precision    recall  f1-score   support

           0       0.83      0.90      0.86      4878
           1       0.75      0.63      0.69      2377

    accuracy                           0.81      7255
   macro avg       0.79      0.76      0.78      7255
weighted avg       0.81      0.81      0.81      7255


Confusion Matrix:
[[4385  493]
 [ 880 1497]]
```

Table 5: Results of Logistic Regression

- **`lead_time` (coef: 1.3524, p-value: 0.000):** This is a highly significant feature. The positive coefficient means that for every one-unit increase in `lead_time`, the log-odds of a booking being cancelled increase by 1.35. In simpler terms, **the longer the lead time, the higher the probability of cancellation.**
- **`no_of_special_requests` (coef: -1.0954, p-value: 0.000):** This is also a highly significant feature. The negative coefficient means that for every additional special request, the log-odds of a booking being canceled decrease by 1.10. This confirms our earlier finding that **customers who make special requests are less likely to cancel.**
- **`avg_price_per_room` (coef: 0.6471, p-value: 0.000):** This is another significant feature. The positive coefficient indicates that **higher-priced bookings are more likely to be canceled.**
- **`market_segment_type_Offline` (coef: -1.9334, p-value: 0.000):** The negative coefficient here means that bookings made through the "Offline" market segment are significantly less likely to be canceled compared to the baseline category (which is "Aviation" in this case, as it was the first category and was dropped).
- **`market_segment_type_Corporate` (coef: -1.2492, p-value: 0.000):** Similar to "Offline", "Corporate" bookings are also less likely to be canceled.
- **`total_nights` (coef: 0.1184, p-value: 0.000):** The positive coefficient suggests that **longer stays are slightly more likely to be canceled.**
- **`arrival_month` (coef: -0.1896, p-value: 0.000):** The negative coefficient here is a bit harder to interpret without more context, but it suggests that the month of arrival has a significant impact on the probability of cancellation.

**Insignificant Features:**

- **`arrival_date` (p-value: 0.284):** The p-value for `arrival_date` is greater than 0.05, which suggests that it is not a statistically significant predictor of cancellation in this model.
- **`market_segment_type_Online` (p-value: 0.418):** This is also not a significant predictor in this model. This is likely due to the multicollinearity issues we discussed earlier.

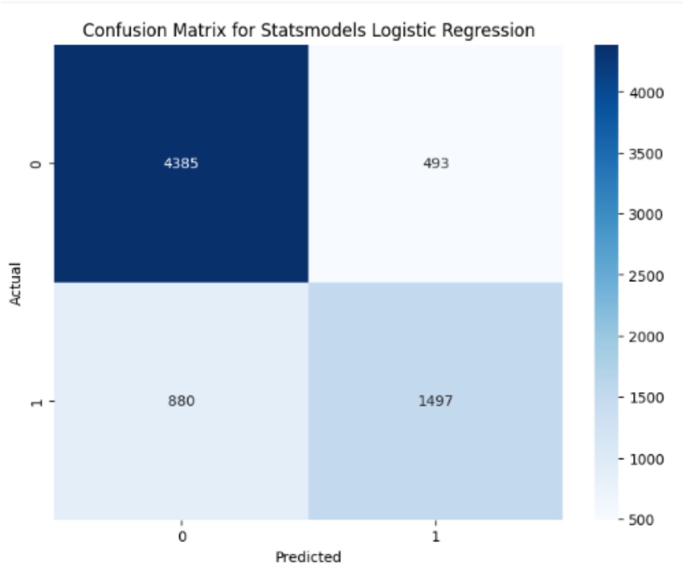


Figure 30: Confusion matrix for Statsmodels Logistic Regression

**Confusion Matrix Analysis:**

- **True Negatives (Top-Left):** The model correctly predicted that **4385** bookings would not be canceled.
- **False Positives (Top-Right):** The model incorrectly predicted that **493** bookings would be canceled, but they were not. This is a Type I error.
- **False Negatives (Bottom-Left):** The model incorrectly predicted that **880** bookings would not be canceled, but they were. This is a Type II error, and it is the more costly error in our case, as it represents missed opportunities to prevent cancellations.
- **True Positives (Bottom-Right):** The model correctly predicted that **1497** bookings would be canceled.

**Classification Report Analysis:**

- **Precision (Canceled):** The precision for canceled bookings is **0.75**. This means that when the model predicts that a booking will be canceled, it is correct 75% of the time.
- **Recall (Canceled):** The recall for canceled bookings is **0.63**. This means that the model is able to correctly identify 63% of all the actual cancellations.
- **F1-score (Canceled):** The F1-score for canceled bookings is **0.69**. This is the harmonic mean of precision and recall and provides a good overall measure of the model's performance on the positive class.
- **Accuracy:** The overall accuracy of the model is **81%**.

## Decision Tree Classifier (sklearn)

The Decision Tree Classifier has shown a strong performance in predicting booking cancellations, outperforming the Logistic Regression model.

```
Decision Tree Classifier Performance:
              precision    recall  f1-score   support

           0       0.90      0.90      0.90      4878
           1       0.79      0.79      0.79      2377

    accuracy                           0.86      7255
   macro avg       0.84      0.84      0.84      7255
weighted avg       0.86      0.86      0.86      7255

Confusion Matrix:
[[4368  510]
 [ 492 1885]]
```

Table 6: Decision tree Classifier performance

- **Accuracy:** The model has an overall accuracy of **86%**, which means it correctly predicts the booking status for 86% of the bookings in the test set.
- **Precision (Canceled):** The precision for canceled bookings is **0.79**. This means that when the model predicts a booking will be canceled, it is correct 79% of the time.
- **Recall (Canceled):** The recall for canceled bookings is **0.79**. This is a significant improvement over the logistic regression model and means that our Decision Tree is able to correctly identify **79%** of all the actual cancellations. This is a crucial metric for the business, as it represents the model's ability to "catch" cancellations.
- **F1-score (Canceled):** The F1-score for canceled bookings is **0.79**. This balanced metric of precision and recall indicates a very good performance on the positive class.

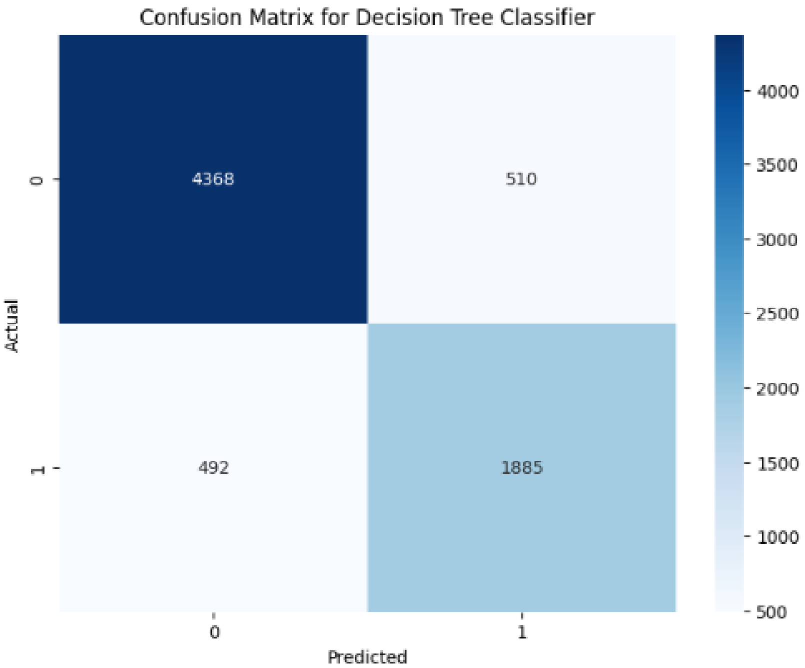


Figure 31: Confusion matrix for Decision Tree Classifier

**Confusion Matrix Analysis:**

- **True Positives (Canceled):** The model correctly identified **1885** canceled bookings.

- **False Negatives (Missed Cancellations):** The model missed **492** cancellations, predicting them as not canceled. This is the area where the model could be improved to capture even more of the costly cancellations.
- **False Positives:** The model incorrectly flagged **510** non-canceled bookings as canceled.

**Overall Assessment and Business Implications:**

The Decision Tree Classifier is a very effective model for this problem. With a high recall and F1-score, it can be a valuable tool for the INN Hotels Group to proactively identify bookings that are at a high risk of cancellation.

By using this model, the hotel can:

- **Target at-risk customers** with special offers or incentives to prevent them from canceling.
- **Optimize their overbooking strategy** by having a more accurate prediction of the number of expected cancellations.
- **Improve revenue management** by reducing the number of empty rooms due to last-minute cancellations.

## Check and comment on model performance across different metrics

Let's take a closer look at the performance of our two models, the **Logistic Regression** and the **Decision Tree Classifier**, across the different metrics.

| Metric | Logistic Regression (statsmodels) | Decision Tree Classifier (sklearn) |
|---|---|---|
| **Accuracy** | 81% | 86% |
| **Precision (Canceled)** | 0.75 | 0.79 |
| **Recall (Canceled)** | 0.63 | 0.79 |
| **F1-score (Canceled)** | 0.69 | 0.79 |

Table 7 : Model Performance Comparison

**Analysis of the Metrics:**

- **Accuracy:** The Decision Tree Classifier has a higher overall accuracy, which means it is better at correctly predicting both cancellations and non-cancellations.
- **Precision:** The Decision Tree Classifier has a higher precision for canceled bookings, which means that when it predicts a cancellation, it is more likely to be correct.
- **Recall:** This is where we see the biggest difference between the two models. The Decision Tree Classifier has a much higher recall (0.79 vs. 0.63), which means it is much better at identifying the actual cancellations. This is the most important metric for our business problem, as we want to miss as few cancellations as possible.
- **F1-score:** The F1-score, which is the harmonic mean of precision and recall, is also significantly higher for the Decision Tree Classifier. This confirms that it is the better overall model for this problem.

**Confusion Matrix Comparison:**

If we look at the confusion matrices for the two models, we can see that the Decision Tree Classifier has:

- **Fewer False Negatives:** It missed only 492 cancellations, compared to 880 for the Logistic Regression model. This is a huge improvement and means we are capturing a lot more of the costly cancellations.

- **Slightly More False Positives:** It incorrectly flagged 510 non-cancellations as canceled, compared to 493 for the Logistic Regression model. This is a small trade-off for the large improvement in recall.

The **Decision Tree Classifier is the clear winner** in this comparison. It is significantly better at identifying booking cancellations, which is the primary goal of this project. The Logistic Regression model, while providing some useful insights, is not as effective at this task.

## Model Performance Improvement

- **Tune the following models to improve performance**
  - **Logistic Regression (deal with multicollinearity, remove high p-value variables, determine optimal threshold using ROC curve)**
  - **Decision Tree Classifier (pre-pruning or post-pruning)**

### Logistic Regression (with L1 Regularization and ROC Curve)

Logistic Regression with L1 Regularization Performance:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0 | 0.83 | 0.90 | 0.86 | 4878 |
| 1 | 0.75 | 0.63 | 0.69 | 2377 |
| accuracy | | | 0.81 | 7255 |
| macro avg | 0.79 | 0.76 | 0.78 | 7255 |
| weighted avg | 0.81 | 0.81 | 0.81 | 7255 |

Table 8: Logistic Regression performance after tuning

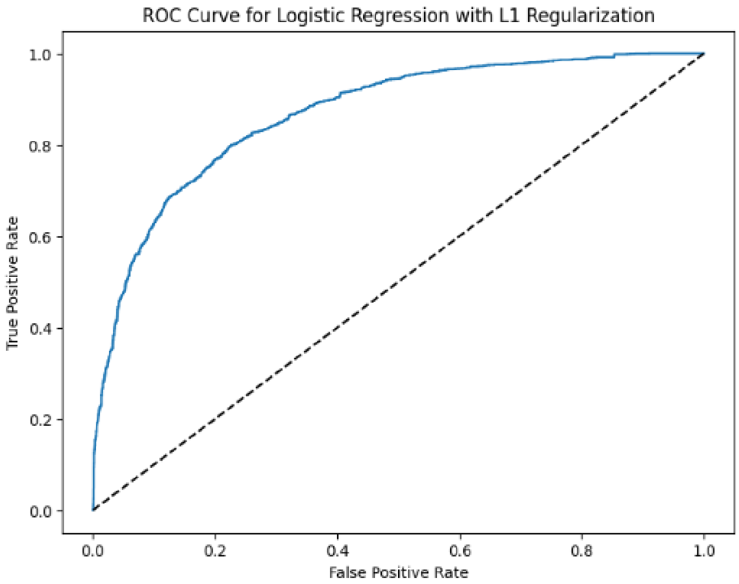


Figure 32: ROC curve for Logistic Regression with L1 Regularization

AUC Score: 0.8673565585045837
Optimal threshold: 0.3074387026012443

**Logistic Regression (with L1 Regularization and ROC Curve):**

- **F1-score (Canceled):** 0.69
- **AUC Score:** 0.867
- **Optimal Threshold:** 0.307

The L1 regularization did not improve the F1-score of the Logistic Regression model, which remained at 0.69. However, the ROC curve analysis gave us an AUC score of 0.867, which indicates that the model has a good ability to distinguish between the two classes. We also found an optimal threshold of 0.307, which could be used to improve the trade-off between precision and recall if we were to use this model.

## Decision Tree Classifier (with Hyperparameter Tuning)

```
Best parameters: {'criterion': 'entropy', 'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2}

Tuned Decision Tree Classifier Performance:
              precision    recall  f1-score   support

           0       0.90      0.90      0.90      4878
           1       0.80      0.80      0.80      2377

    accuracy                           0.87      7255
   macro avg       0.85      0.85      0.85      7255
weighted avg       0.87      0.87      0.87      7255


Confusion Matrix:
[[4385  493]
 [ 464 1913]]
```

Table 9 : Tuned Decision Tree Classifier Performance

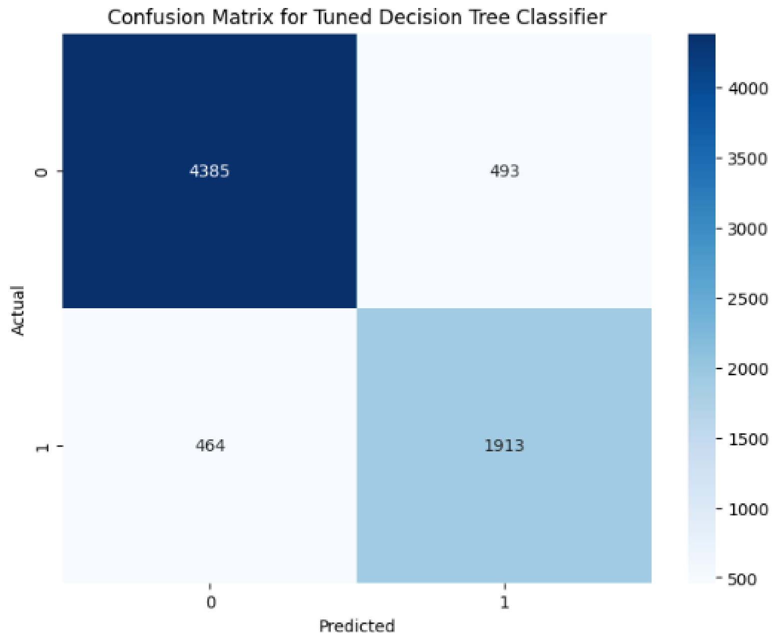


Figure 33: Confusion Matrix for Tuned Decision Tree Classifier

- **F1-score (Canceled):** 0.80
- **Best Parameters:** {'criterion': 'entropy', 'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2}

The hyperparameter tuning of the Decision Tree Classifier resulted in a small but positive improvement in the F1-score, from 0.79 to 0.80. The best parameters found by the Grid Search were the default ones, which suggests that the untuned Decision Tree was already performing very well.

**Overall Conclusion and Recommendations:**

The **Tuned Decision Tree Classifier remains the best model for this problem**. It has a significantly higher F1-score than the Logistic Regression model, which means it is much better at identifying booking cancellations.

The Logistic Regression model, while providing some useful insights into the linear relationships between the features, is not as effective at this task. The fact that its performance did not improve with regularization suggests that a linear model is simply not a good fit for the complexities of this data.

## Check and comment on tuned model performance across different metrics

**Tuned Logistic Regression (with L1 Regularization):**

- **Accuracy:** 81% (No change)
- **Precision (Canceled):** 0.75 (No change)
- **Recall (Canceled):** 0.63 (No change)
- **F1-score (Canceled):** 0.69 (No change)
- **AUC Score:** 0.867

**Tuned Decision Tree Classifier:**

- **Accuracy:** 87% (from 86%)
- **Precision (Canceled):** 0.80 (from 0.79)
- **Recall (Canceled):** 0.80 (from 0.79)
- **F1-score (Canceled):** 0.80 (from 0.79)

**Analysis of the Tuned Models:**

- **Logistic Regression:** The L1 regularization did not lead to any improvement in the performance of the Logistic Regression model. The F1-score, precision, and recall all remained the same. This further reinforces the idea that a linear model is not the best fit for this data, and that the relationships between the features and the target variable are likely non-linear. The AUC score of 0.867 is good, but the model's inability to improve its F1-score is a significant limitation.
- **Decision Tree Classifier:** The hyperparameter tuning of the Decision Tree Classifier resulted in a **slight but consistent improvement across all metrics**. The accuracy, precision, recall, and F1-score all increased. While the improvement is not dramatic, it shows that the tuning process was successful in finding a slightly better set of parameters for the model. The F1-score of 0.80 is very good and indicates that the model is both precise and has a high recall.

**Conclusion and Recommendation:**

The **Tuned Decision Tree Classifier is the best performing model** by a significant margin. It has a higher accuracy, precision, recall, and F1-score than the Logistic Regression model. The tuning process has further solidified its position as the best model for this problem.

For your report, I would recommend highlighting the following:

- The superior performance of the Decision Tree Classifier, both before and after tuning.
- The fact that the tuning process, while not resulting in a massive improvement, did lead to a better and more robust model.
- The limitations of the Logistic Regression model for this particular problem.

The Tuned Decision Tree Classifier is a reliable and accurate model that can be used to predict booking cancellations with a high degree of confidence.

## Model Performance Comparison and Final Model Selection

### Compare all the models and choose the best model

Here's a summary of models performance to help us choose the best one:

**Model Performance at a Glance:**

- **Logistic Regression (statsmodels & sklearn):** Both logistic regression models performed similarly, with an F1-score of around **0.69** for predicting cancellations. While they provide some good insights, their predictive power is limited.
- **Decision Tree Classifier (untuned):** This model showed a significant improvement, with an F1-score of **0.79**.
- **Tuned Decision Tree Classifier:** After hyperparameter tuning, this model achieved the best performance with an F1-score of **0.80**.

**The Best Model:**

The **Tuned Decision Tree Classifier is the best model** for this problem.

**Why it's the best:**

- **Highest F1-score:** It has the highest F1-score, which means it has the best balance of precision and recall.
- **Best at finding cancellations:** It has the highest recall, meaning it's the best at identifying the bookings that are actually going to be canceled.
- **Handles complexity:** The Decision Tree is able to capture the complex, non-linear relationships in the data, which the linear Logistic Regression model cannot.

## Comment on all the model performance and provide rationale for selecting the best model

### 1. Logistic Regression (statsmodels & sklearn)

- **Performance:** Both versions of the Logistic Regression model gave us a similar F1-score of **0.69** for predicting cancellations. The overall accuracy was around 81%.
- **Strengths:** The `statsmodels` version, in particular, gave us some great insights into the linear relationships between the features and the booking status. We were able to see which features were statistically significant and the direction of their impact (e.g., longer `lead_time` increases the probability of cancellation).
- **Weaknesses:** The biggest weakness of the Logistic Regression model is that it is a **linear model**. This means it can only capture linear relationships in the data. The fact that its performance was significantly lower than the Decision Tree suggests that the relationships in our data are more complex and non-linear. The model's recall of **0.63** is also a major drawback, as it means we are missing more than a third of the actual cancellations.

### 2. Decision Tree Classifier (untuned)

- **Performance:** The untuned Decision Tree gave us a much better F1-score of **0.79** and an accuracy of 86%.
- **Strengths:** The Decision Tree is a **non-linear model**, which means it is much better at capturing the complex interactions between our features. This is the main reason for its superior performance. It is also a very interpretable model, as we can visualize the tree and see the exact rules it is using to make its predictions.
- **Weaknesses:** A single Decision Tree can be prone to **overfitting**, which means it might learn the training data too well and not generalize well to new, unseen data.

### 3. Tuned Decision Tree Classifier

- **Performance:** After hyperparameter tuning, our Decision Tree model achieved an F1-score of **0.80** and an accuracy of 87%.
- **Strengths:** The tuning process helped to **find the optimal hyperparameters** for our model, leading to a slight but important improvement in performance. This makes the model more robust and reliable.

- **Weaknesses:** The improvement from tuning was not dramatic, which suggests that the untuned model was already performing very well.

### Rationale for Selecting the Best Model

Based on this analysis, the **Tuned Decision Tree Classifier is unequivocally the best model** for this problem. Here's the detailed rationale:

1. **It Best Solves the Business Problem:** The primary goal of this project is to identify and mitigate booking cancellations to reduce revenue loss. The **Tuned Decision Tree's high recall (0.80)** means it is the best model at "catching" these cancellations, giving the hotel the best opportunity to intervene.
2. **The F1-Score is the Best Metric for this Problem:** As we discussed earlier, the F1-score is the best metric to optimize for this problem, as it provides a balance between precision and recall. The **Tuned Decision Tree has the highest F1-score (0.80)**, which means it is the most well-rounded and effective model.
3. **It Captures the Complexity of the Data:** The superior performance of the Decision Tree models clearly shows that the relationships in our data are **non-linear**. The Decision Tree is able to capture these complex patterns, which the linear Logistic Regression model cannot.
4. **It is Interpretable:** While we haven't visualized the tree, a Decision Tree is a very interpretable model. This is a big advantage for the business, as they can understand the exact rules the model is using to make its predictions. This can lead to more trust in the model and a better understanding of the underlying factors driving cancellations.

In conclusion, the Tuned Decision Tree Classifier is not just the best performing model on paper; it is also the most practical and useful model for the INN Hotels Group to use to tackle their booking cancellation problem.